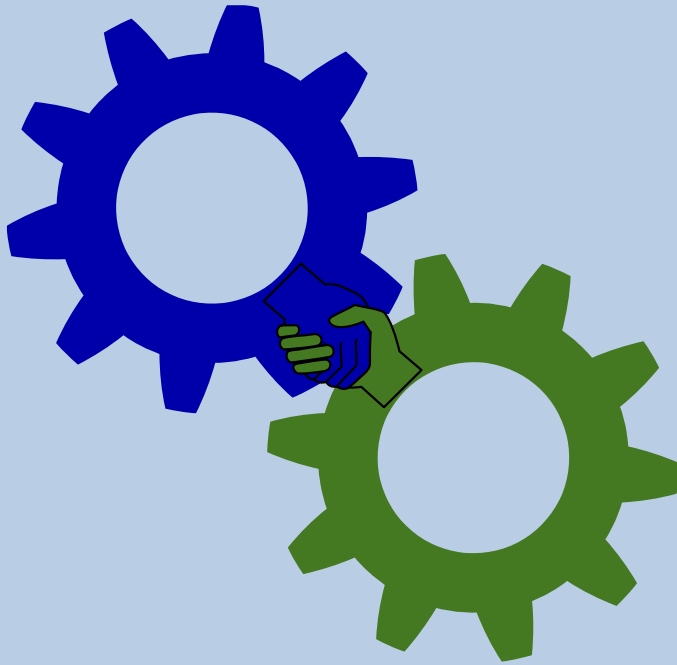




Complex (Nonlinear) Negotiation

Mark Klein

<http://cci.mit.edu/klein/>



**Massachusetts
Institute of
Technology**

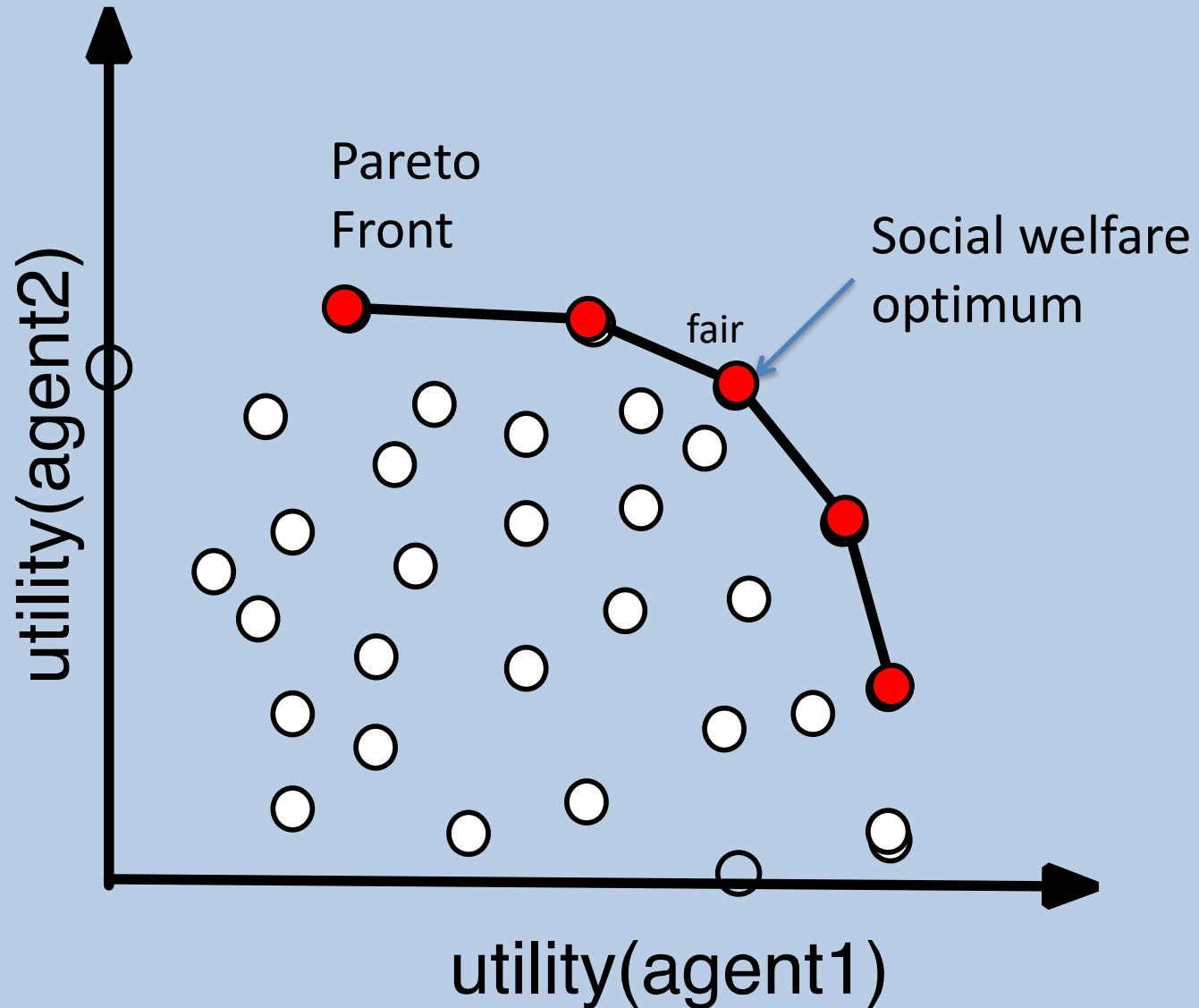
Overview

- What is (complex) negotiation?
- Where is it useful?
- Limitations with state of the art
- Solutions we have developed
- Future directions

What Is Negotiation?

how self-interested parties come to agreements about issues of shared interest

It Should Achieve Optimal Agreements



... With Self-Interested Parties

- Privacy
 - minimize information revelation
- Manipulation
 - Agents may want to lie/cheat to win better outcomes
 - But this has several bad effects
 - Usually leads to prisoner's dilemma that hurts everybody
 - Creates expensive “agent arms race”
 - Discourages new entrants (who fear sneaky competitors)
- Strategy
 - Design mechanisms where the agent's *dominant* (best) strategy is to tell the truth, allowing simple agents and fair, social/pareto-optimal outcomes

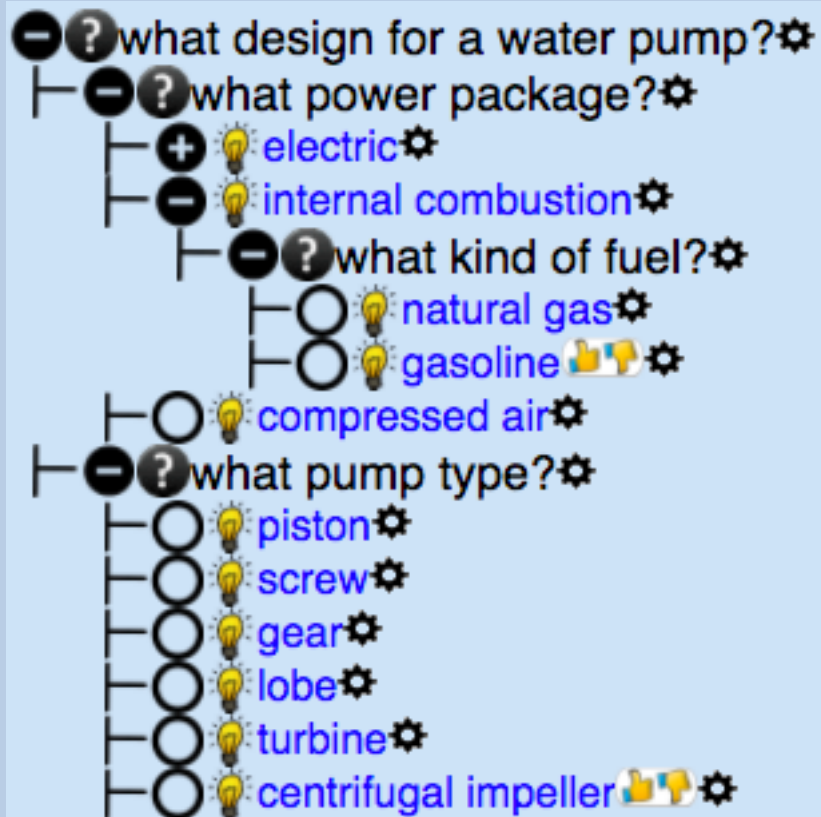
... and Remain Tractable Even With Complex Decisions

- Complexity from
 - Many issues
 - Many agents
 - Many issue *interdependencies*
- Amount of utility elicitation
 - can be expensive and/or a burden for humans
- Amount of Computation
 - Agent (e.g. choosing bids, modeling other agents)
 - Mediator (e.g. winner determination)

Where Is It Useful?

Interdependent Decisions with Multiple
Stakeholders

Collaborative Design



Multiple interdependent decisions

- designs w/multiple parts
- policies w/multiple clauses
- shop floors
- transport networks

SCM Coordination

Supply chain = factories/suppliers/customers, connected by transport links

Pain Points

- *Slow and difficult* to design multi-company supply chains, so we use non-optimal plans, and adapt poorly to changes

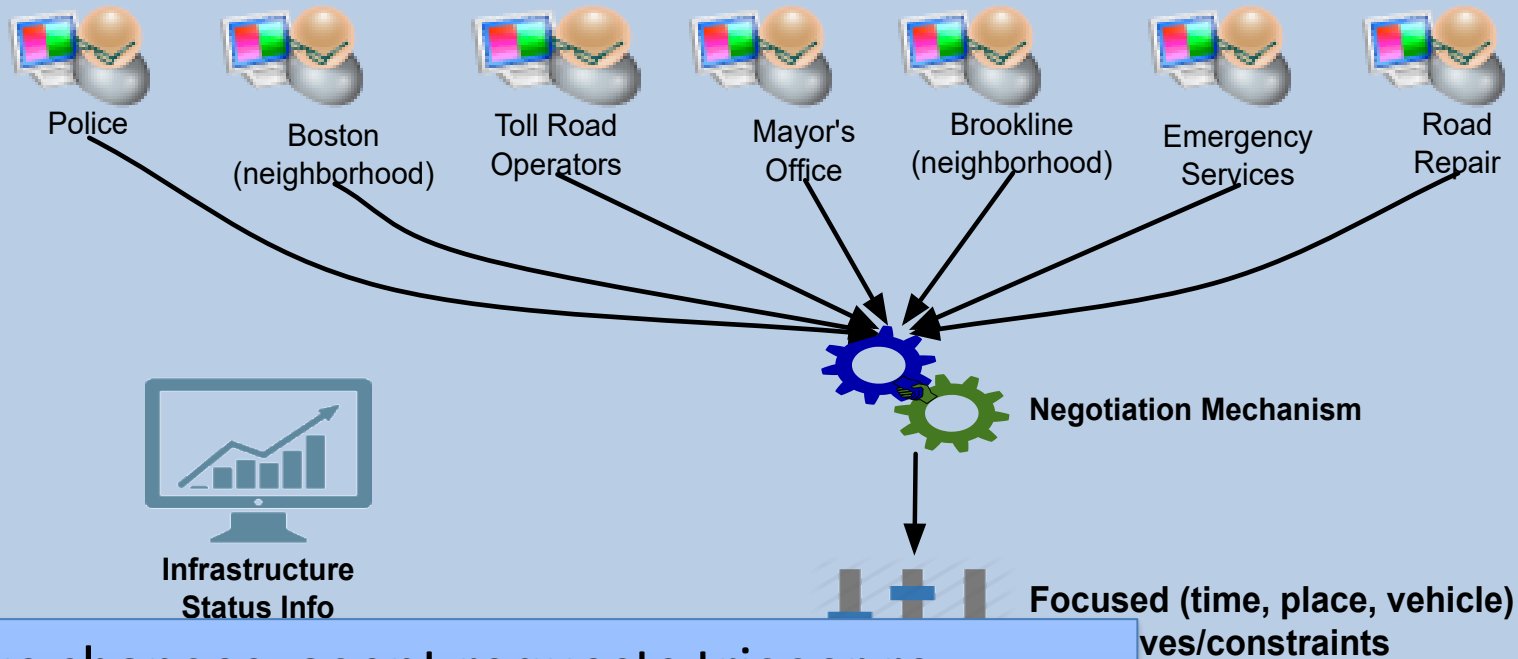
Solution Concept

- All players negotiate, multi-laterally, a single coherent set of supply chain commitments (who ships/creates what, when ...)

Benefits

- quickly design *tailored* supply chains
- quickly respond to contingencies by changing SCM design

Transport Networks



Status changes, agent requests trigger re-negotiation

- e.g. I'll pay 50% more to save 30 minutes from X to Y
- e.g. traffic accident
- e.g. baseball game is ending
- e.g. emergency vehicles need expedited route

ves/constraints

s (incentives

and 6 am

between 6 and 8 am

ake resource utilization

decisions in this policy context
e.g. using "smart GPS"

Limitations of State of the Art

Current Protocols do Fine w/*Simple* Decisions

One or a few *independent* issues e.g. price

Utilities are calculated as *the weighted sum* of values for each contract issue

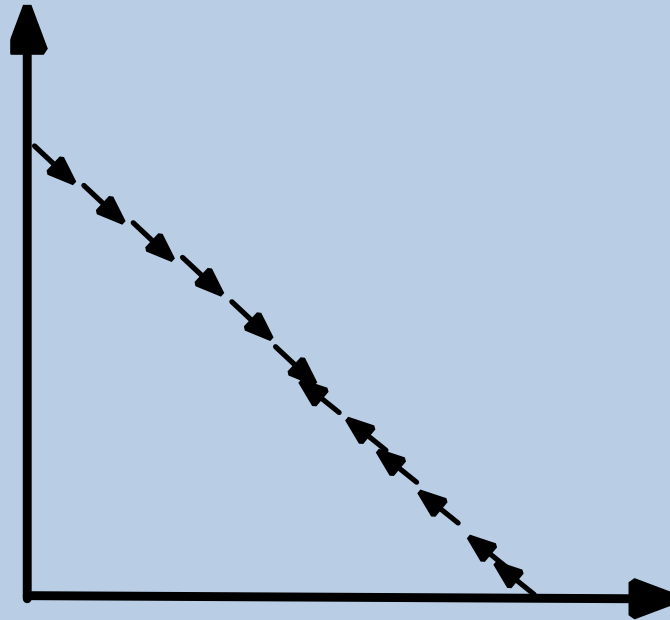
$$U(X_1, X_2, \dots, X_n) = \sum_{i=1}^n w_i * U_i(X_i)$$

	Issue1	Issue2	Issue3	Issue4	Issue5
Agent1	0.33	0.81	-0.74	-0.12	0
Agent2	-0.22	0.31	0.55	-0.81	0.41

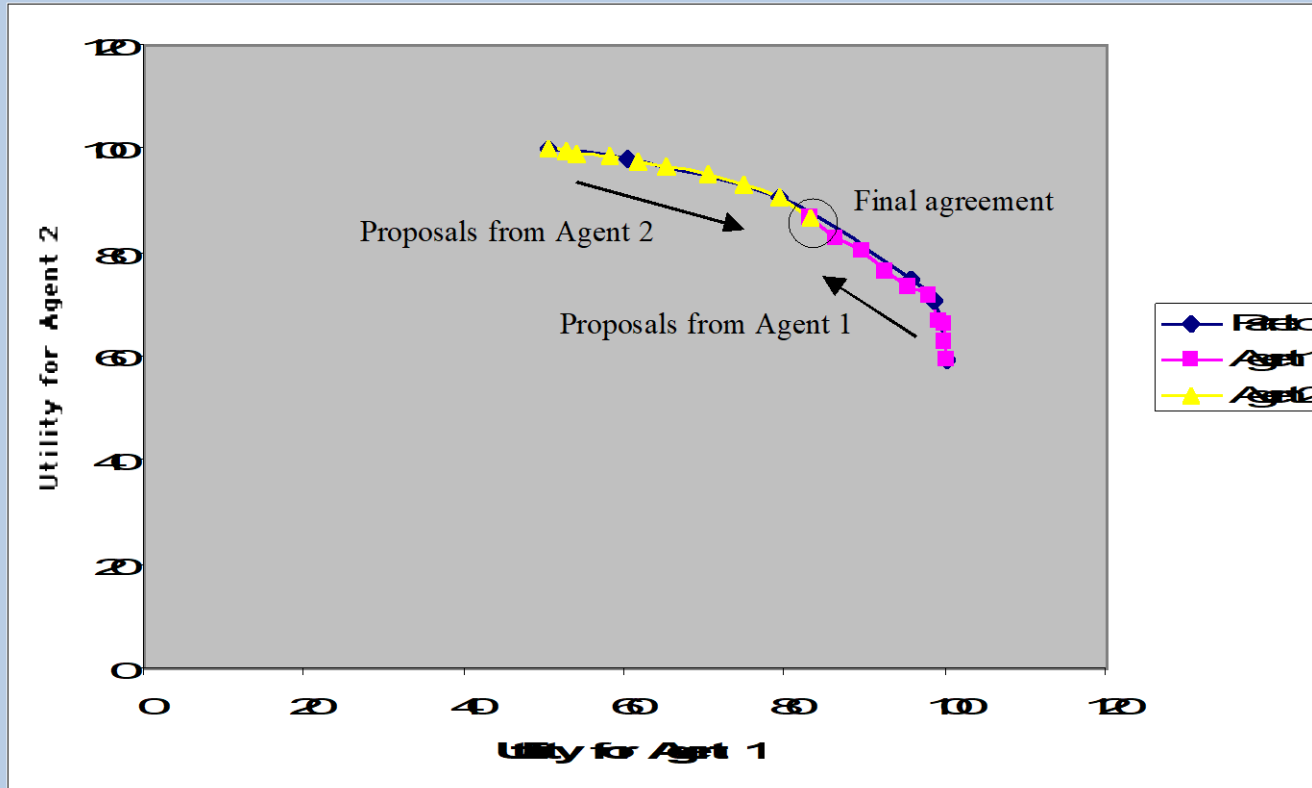
The focus of the vast majority (90+%) of research in negotiation protocols, driven by electronic commerce

Bargaining Protocol

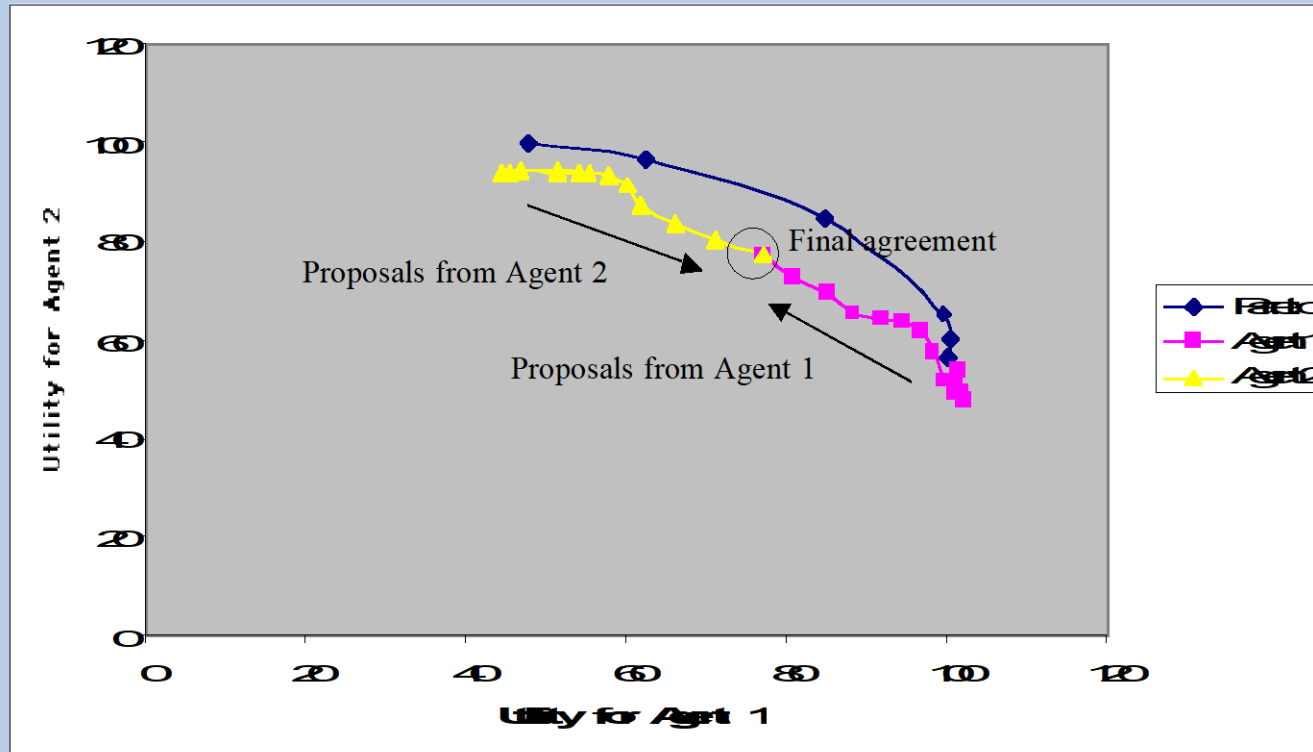
- Each agent starts by proposing their **ideal** contract
- And gradually **modifies** it to be closer to opponent's proposal
- Conceding utility as **slowly** as they can
- Until the two proposals **match**



Optimal for Simple Decisions

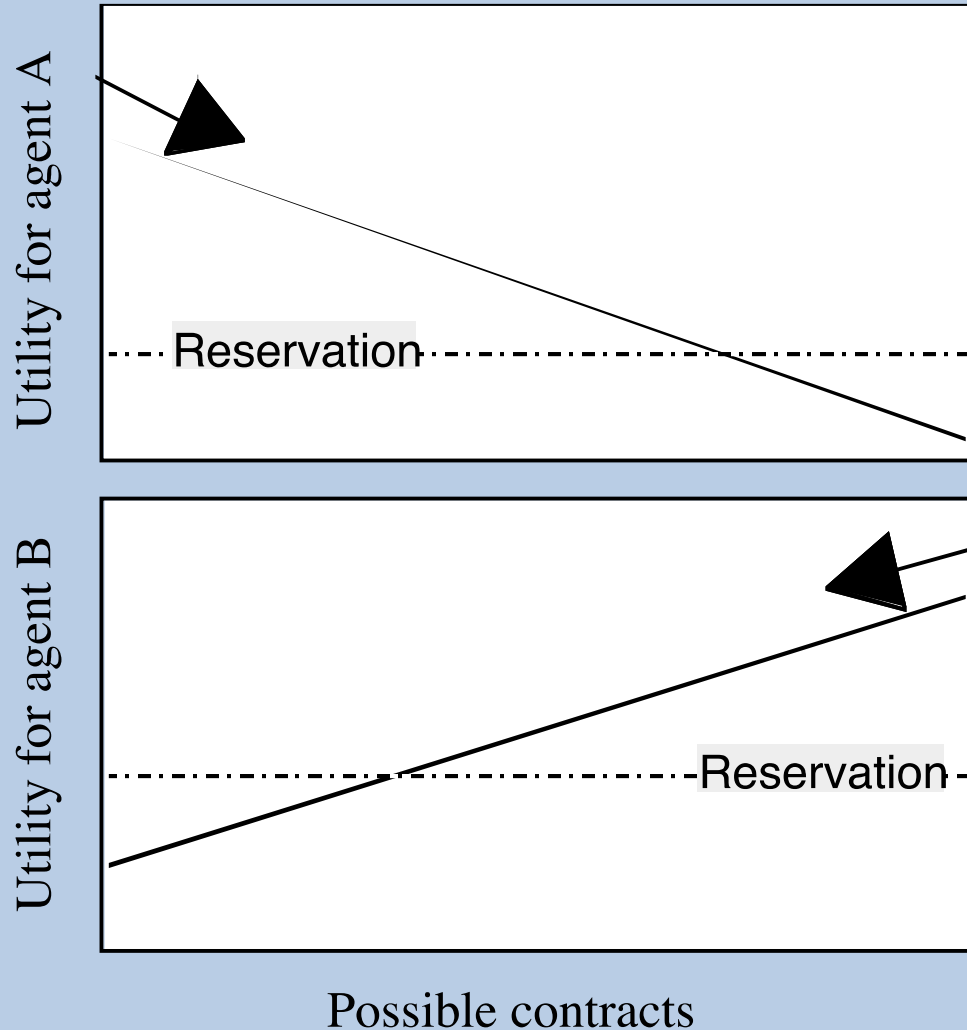


Fails for *Complex** Decisions



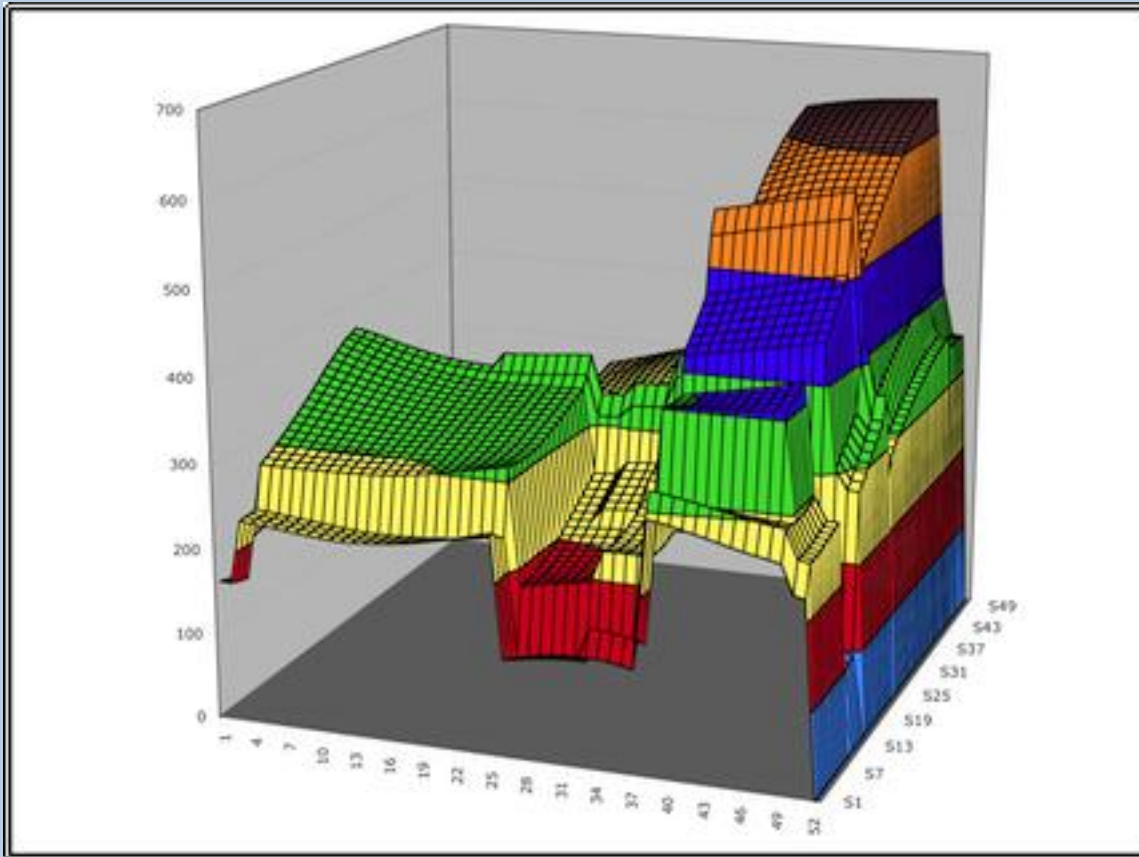
* Many *interdependent* issues

Simple Decisions Have Linear* Utilities



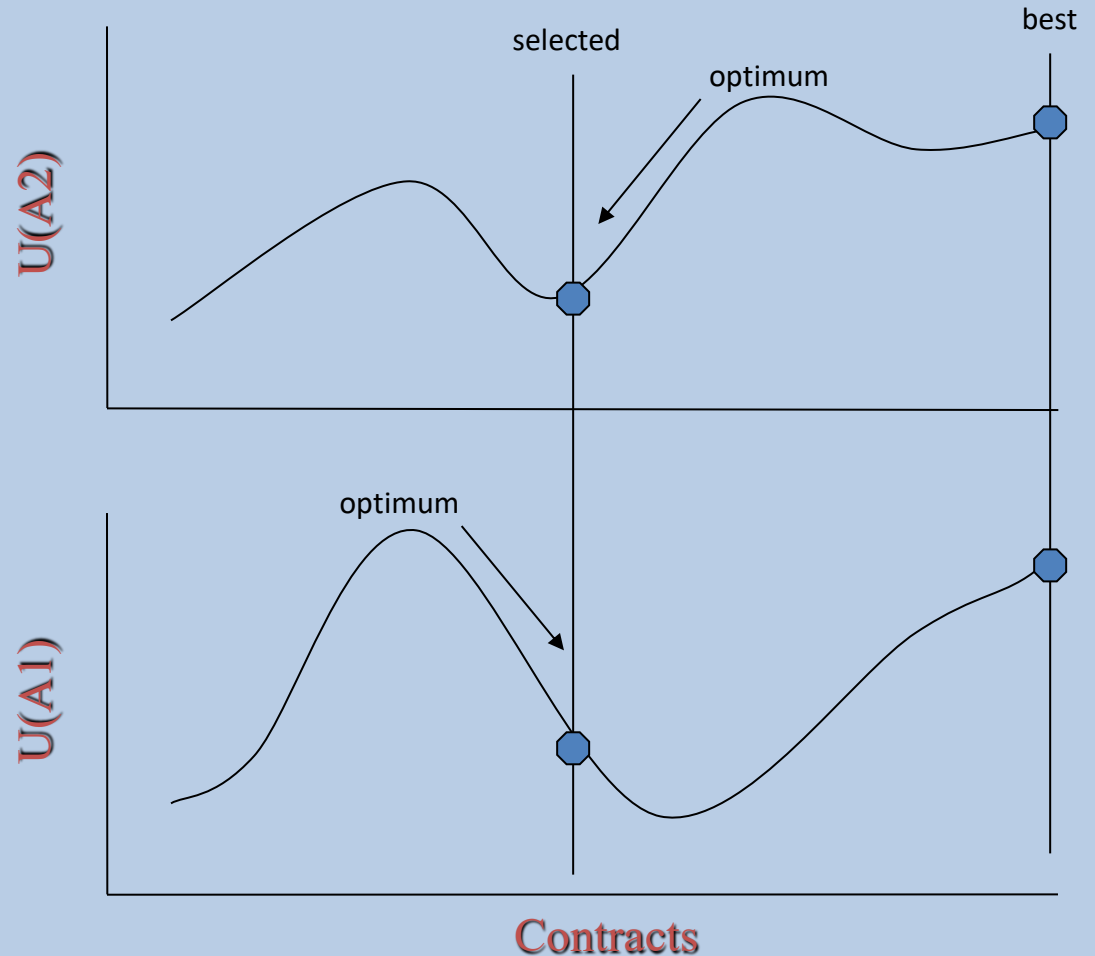
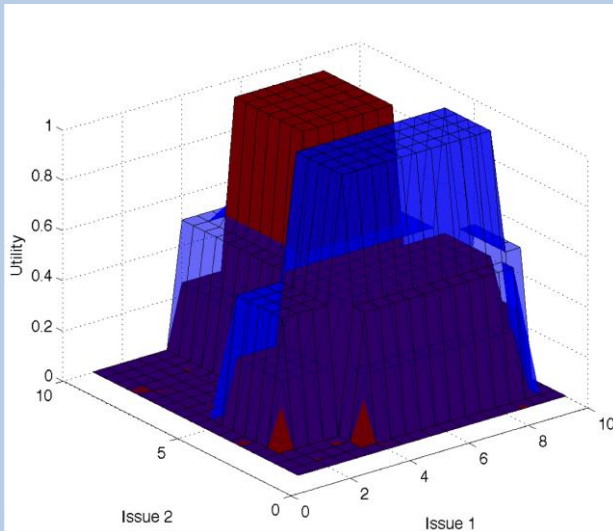
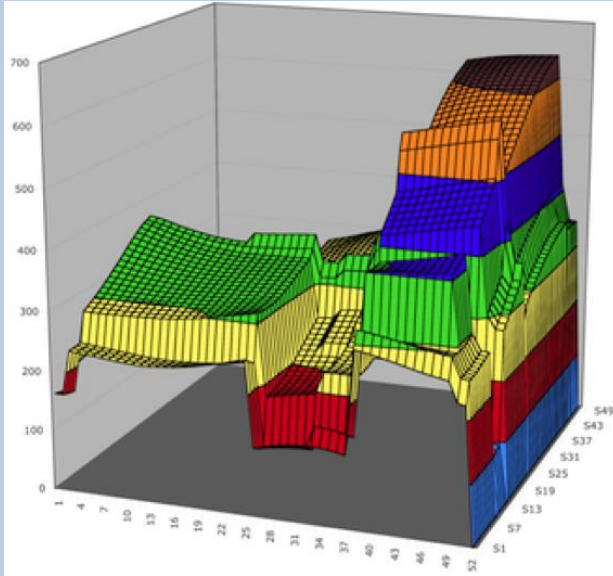
* Or monotonic/concave utilities: single optima

Complex Negotiations => Bumpy Utilities



Utility = a function of
issue value *combinations*
(not just individual
values)

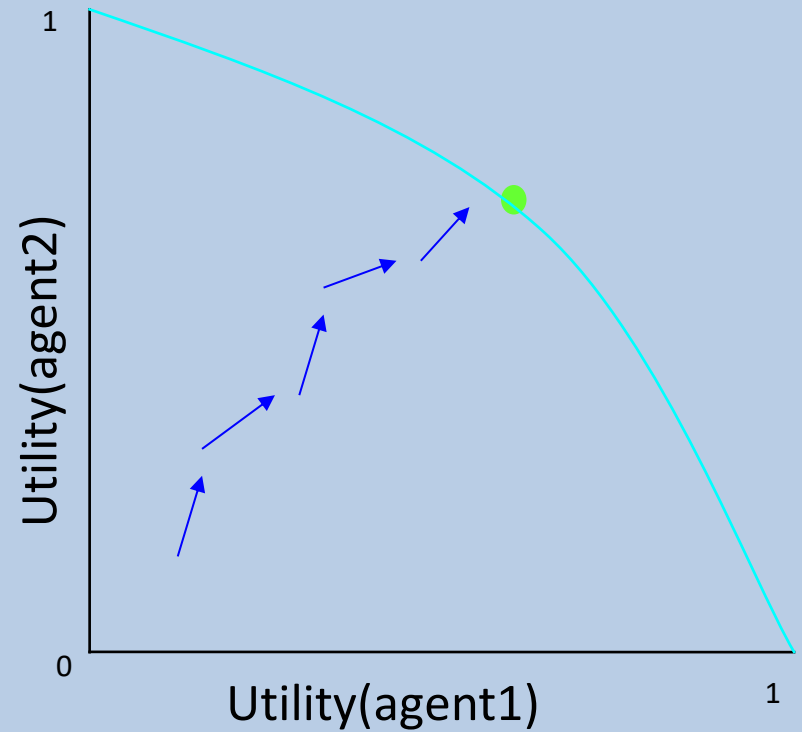
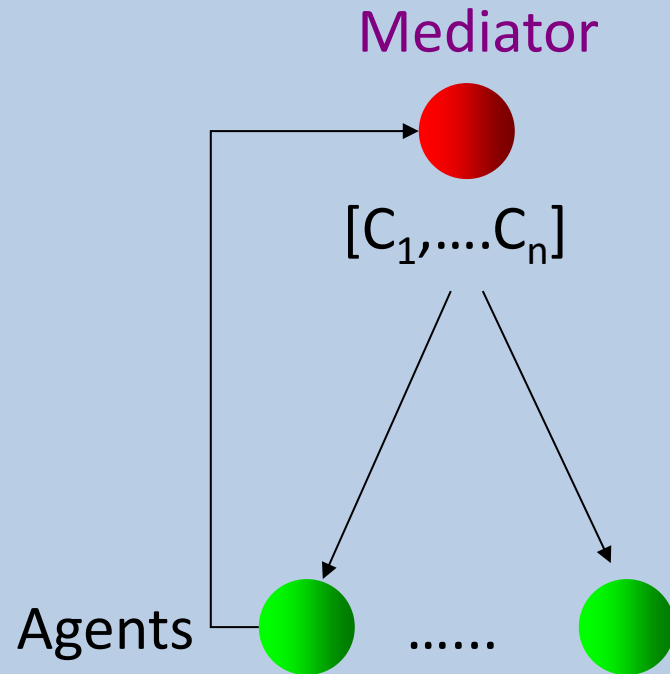
Local sub-optima traps



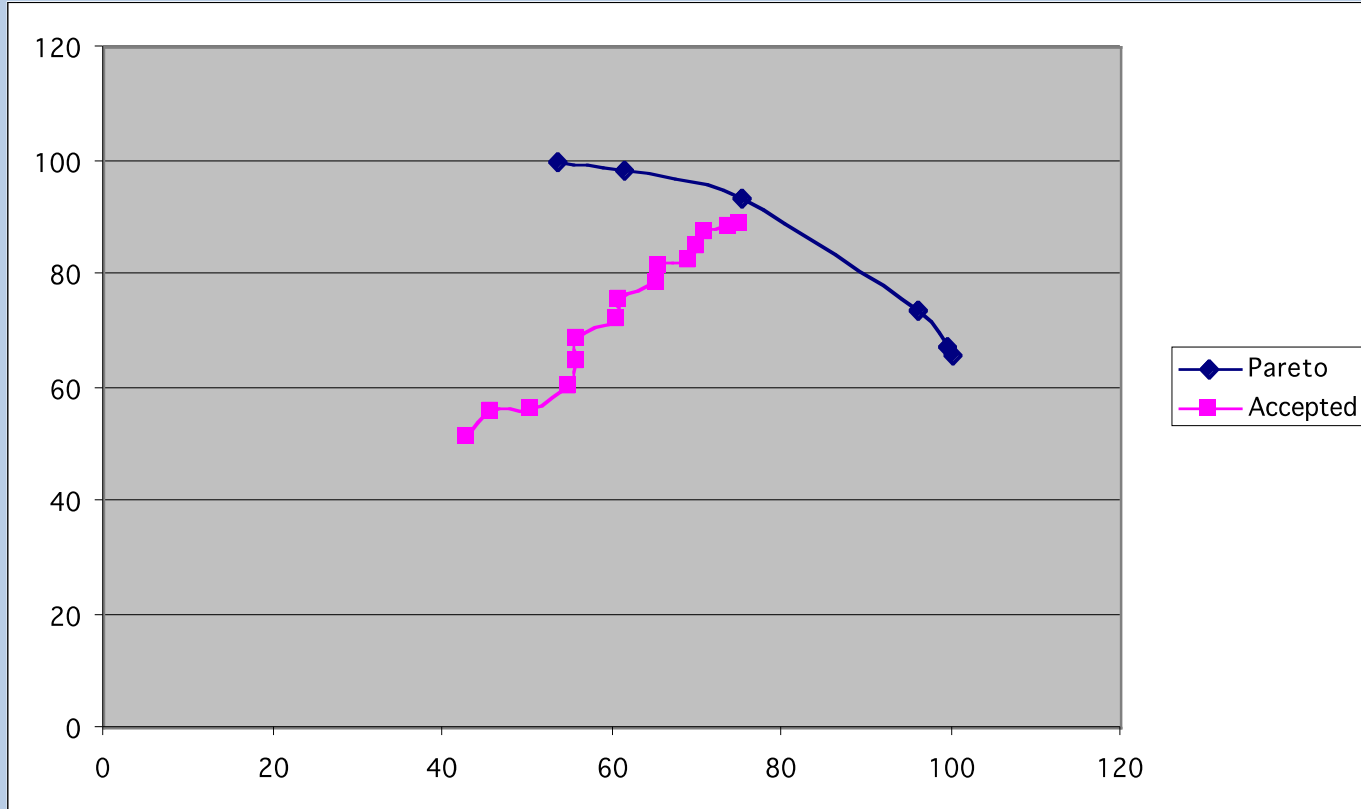
Bargaining can get you stuck in suboptimal agreements

Solutions we have developed:
integrating nonlinear optimization w/
mechanism design

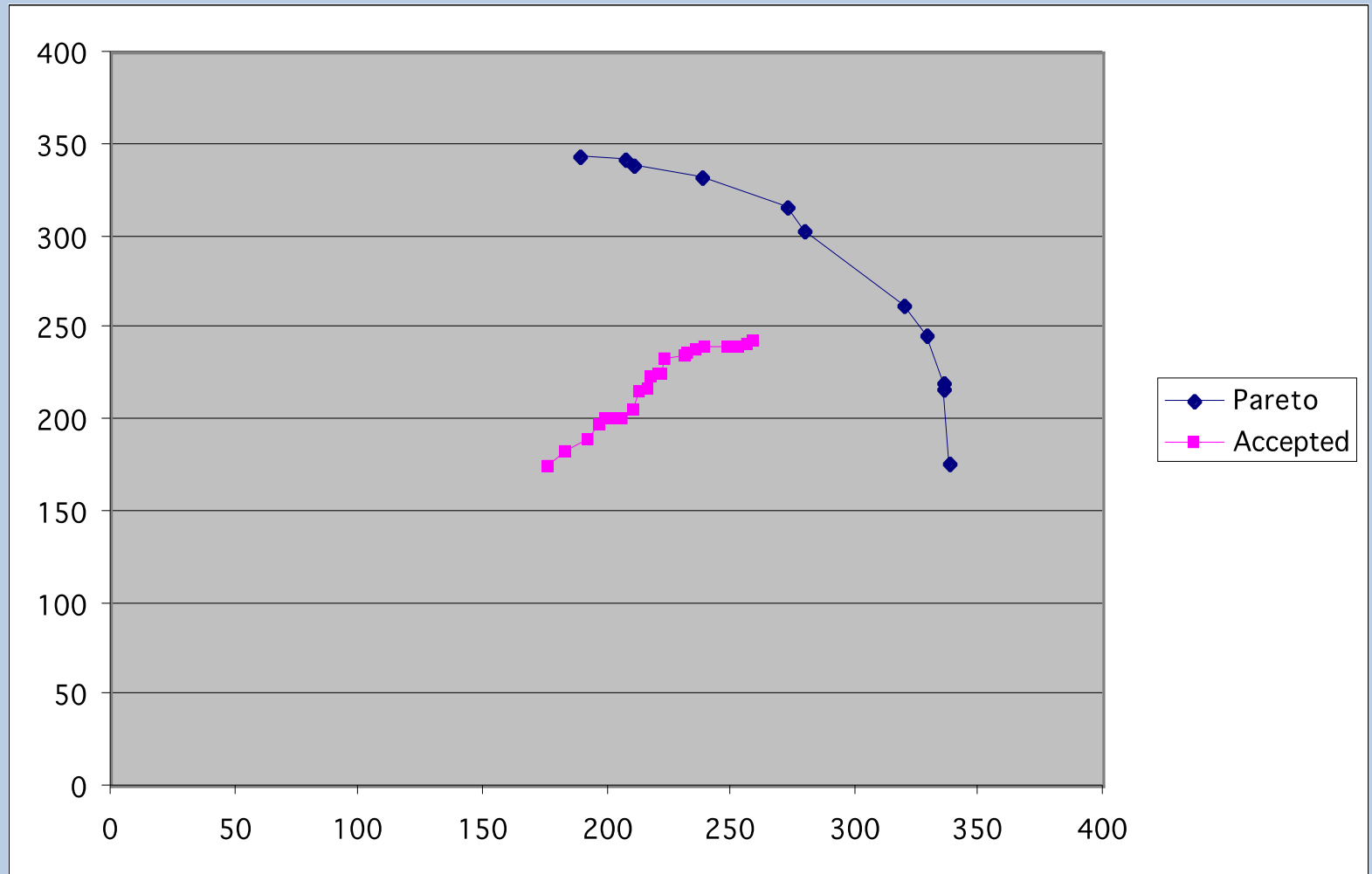
Single Text Negotiation Protocol



Optimal for Simple Decisions

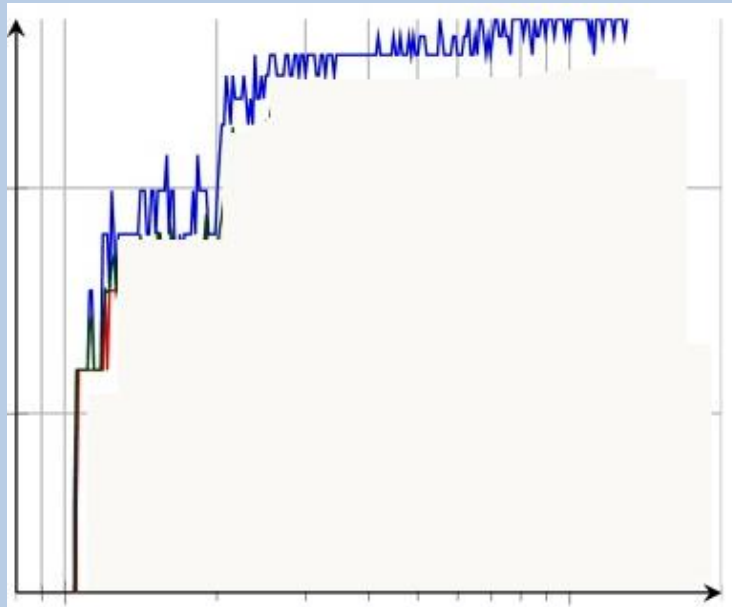


Poor Results in Complex Negotiations



Simulated Annealing

Give agents a time-decreasing willingness to provisionally accept changes that *reduce* their utility, in the hope of jumping past local optima to find a better solution at the end



Single Text w/Annealing Mediator

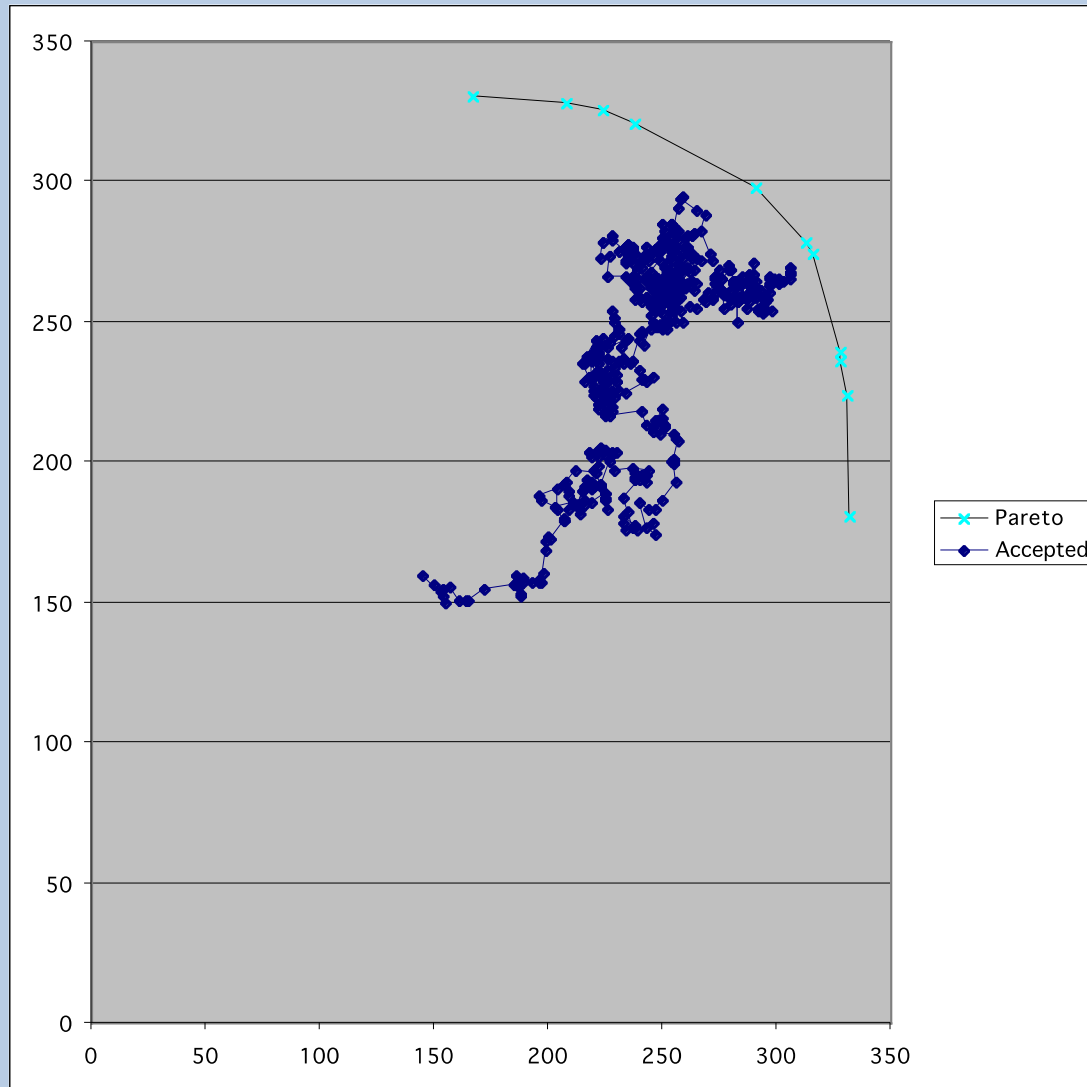
Agents vote on proposed agreement

- Strong accept = 2, weak accept = 1
- Strong reject = -2, weak reject = -1

Mediator stochastically ignores rejections w/time-decreasing probability

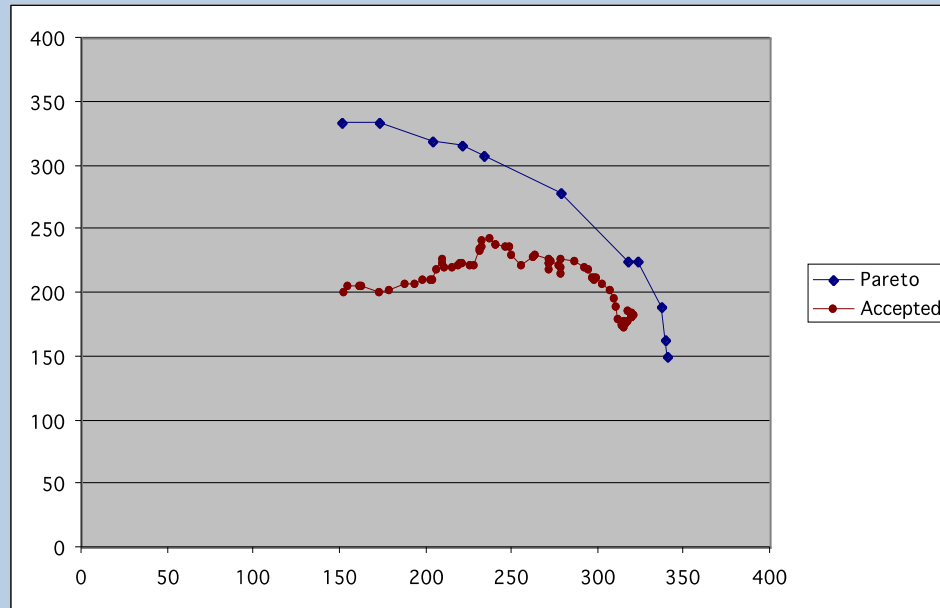
$$P(\text{Accept}) = \begin{cases} 1 & \text{if } \text{sum}(\text{votes}) \geq 0 \\ e^{\text{sum}(\text{votes})/T} & \text{otherwise} \end{cases}$$

Protocol is Near-Optimal



Incentive Compatibility

What if one agent exaggerates and the other doesn't?



The exaggerator wins,
and the truth-teller loses

		ex	truth
ex	ex	84 73/74	84 99/51
	truth	84 51/99	98 84/84

A **prisoner's dilemma game**

truth = collaborate

exaggerate = defect

Cooperation is ***NOT*** a dominant
strategy => **lose-lose** outcomes

Inter-Negotiation Tit-for-Tat

If agents have *many repeated negotiations*,
start as an truth-teller (cooperator),
and **switch** to exaggeration if the other exaggerates,
then cooperation becomes the **dominant strategy**

But what if agents **don't** have repeated negotiations?

Clarke Tax

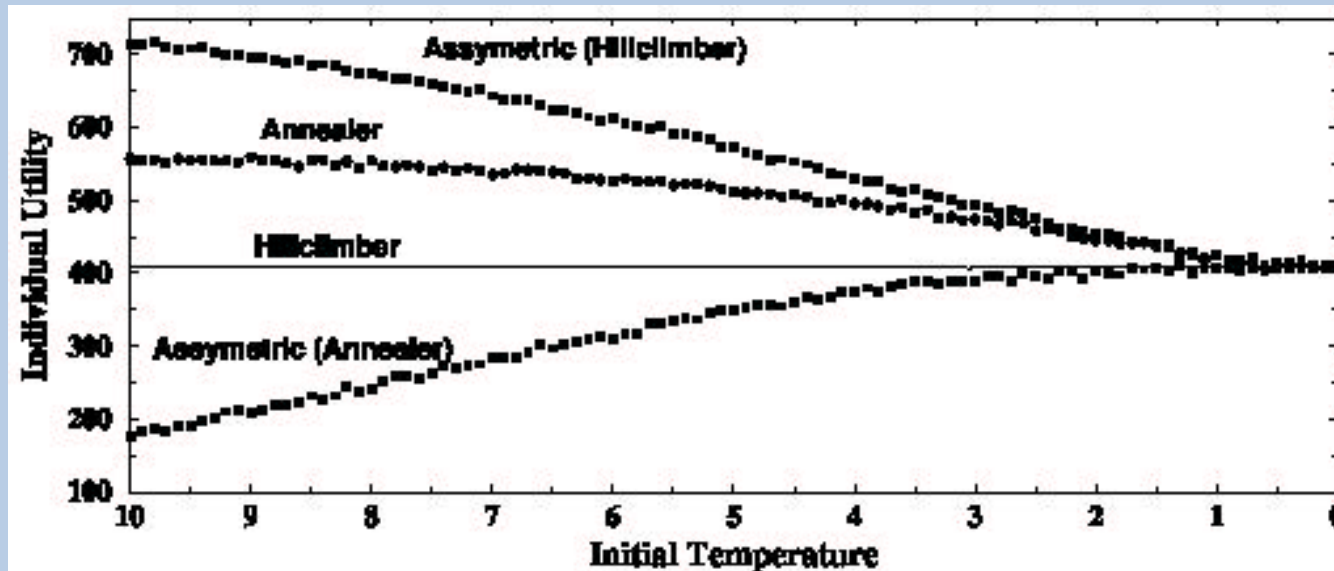
- A way to make truth-telling the rational strategy for an agent
- Every agent pays for *its impact on the other agents*

Agent	Other winners w/agent	Other winners w/o agent	Impact on others (tax)
A1	A3: 2	A2: 6	-4
A2	A1: 5 A3: 2	A1: 5 A3: 2	0
A3	A2: 5	A3: 6	-1

- So agents only bid if it's *honestly* worth it for them

“Cold” Annealing

Let's say the annealer starts at a lower temperature, so that it can not be dragged as far from its own optimum



If the annealer agent starts at a low enough temperature ($T_0 = 2$ in this case), the conceder's penalty is in fact eliminated, but the hillclimber still wins and social welfare is only slightly better than that achieved by two hill climbers.

Parity-Preserving Annealing Mediator

Key trick

- If agent1 wins and agent2 loses in a round, agent1 must pay a **token** to agent2. If agent1 has no tokens, round is annulled.
- Agents have a limited budget of tokens, so they make strong votes if it **truly** benefits them
- like a Clarke tax, except that tax is on *the ability to influence subsequent rounds* (rather than a monetary tax)

Result

- Outcomes are 98% optimal
- Truth-telling is *empirically* the individually rational strategy (based on simulations, not theoretical proof)

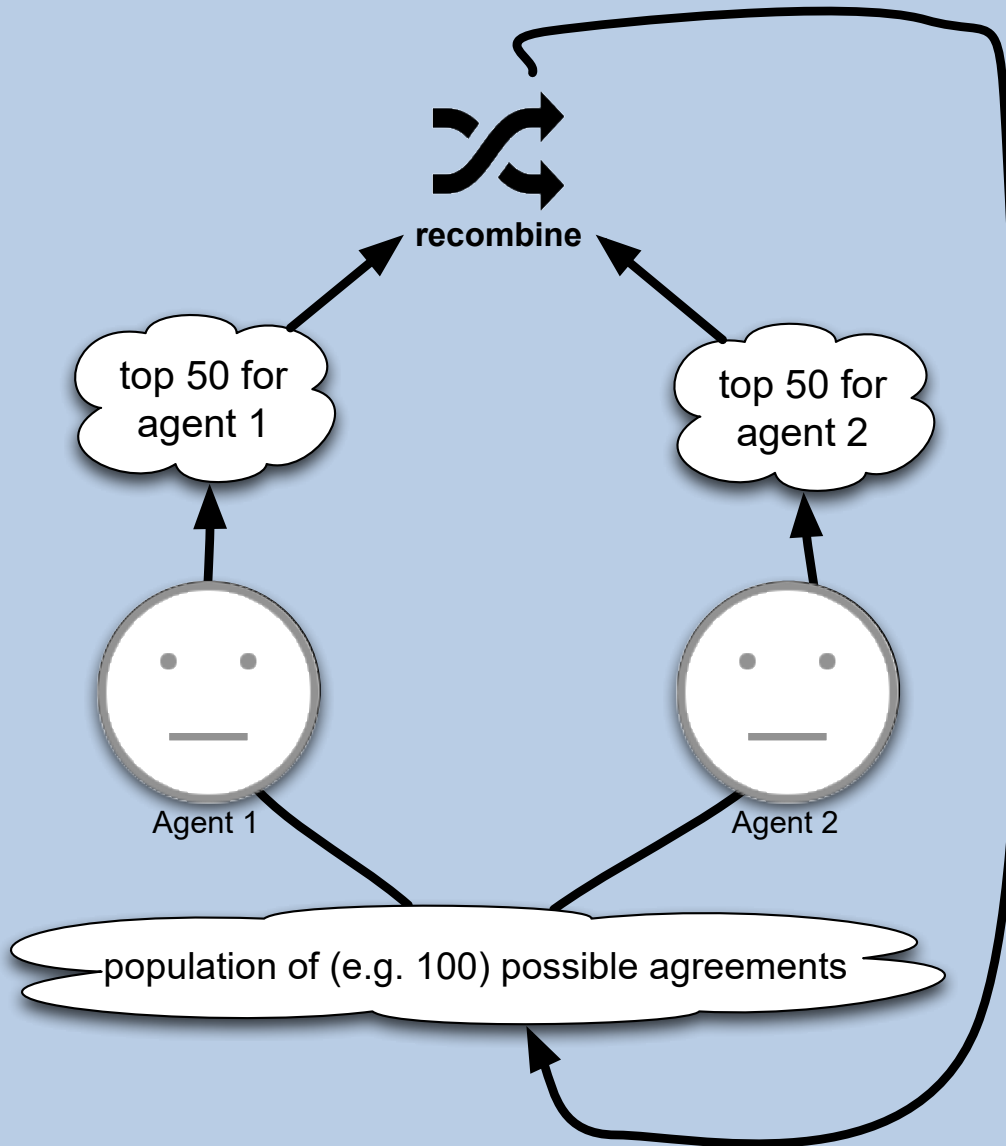
Accomplishments

- Near-optimal outcomes for complex contracts (10^{30} contracts, 100's of constraints) with two agents
- *Empirical* demonstration of incentive compatibility
- *Moderate* information revelation (2500 pair comparisons per negotiation) for mediated SA

The Scalability Challenge

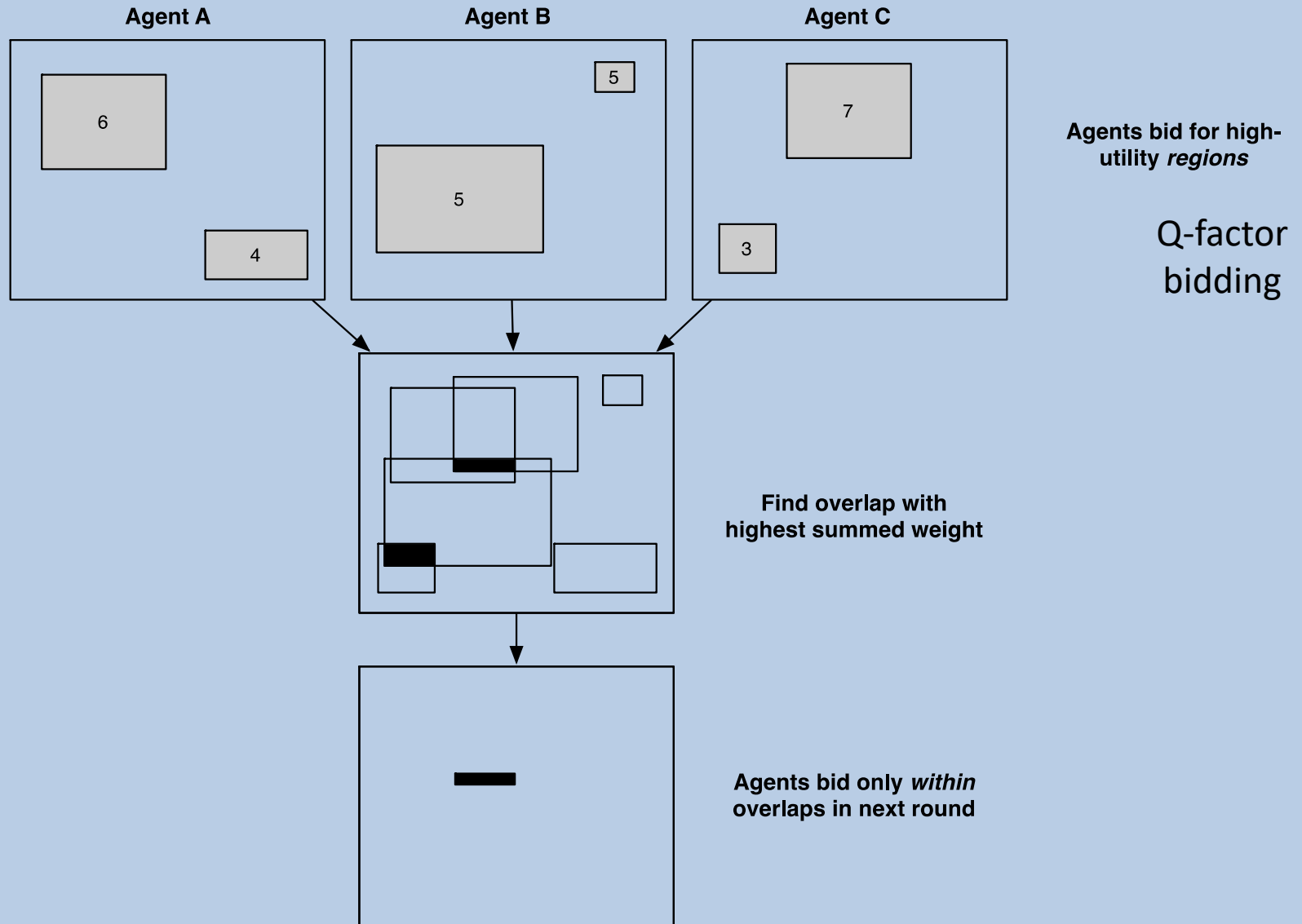
- Computational tractability
 - 10' s to 100' s of agents
 - 100' s to 1000' s of issues
- Amount of preference revelation
- Suitability for inter-human negotiation
 - Cognitive load
 - # of rounds

Mediated Genetic Algorithm Protocol



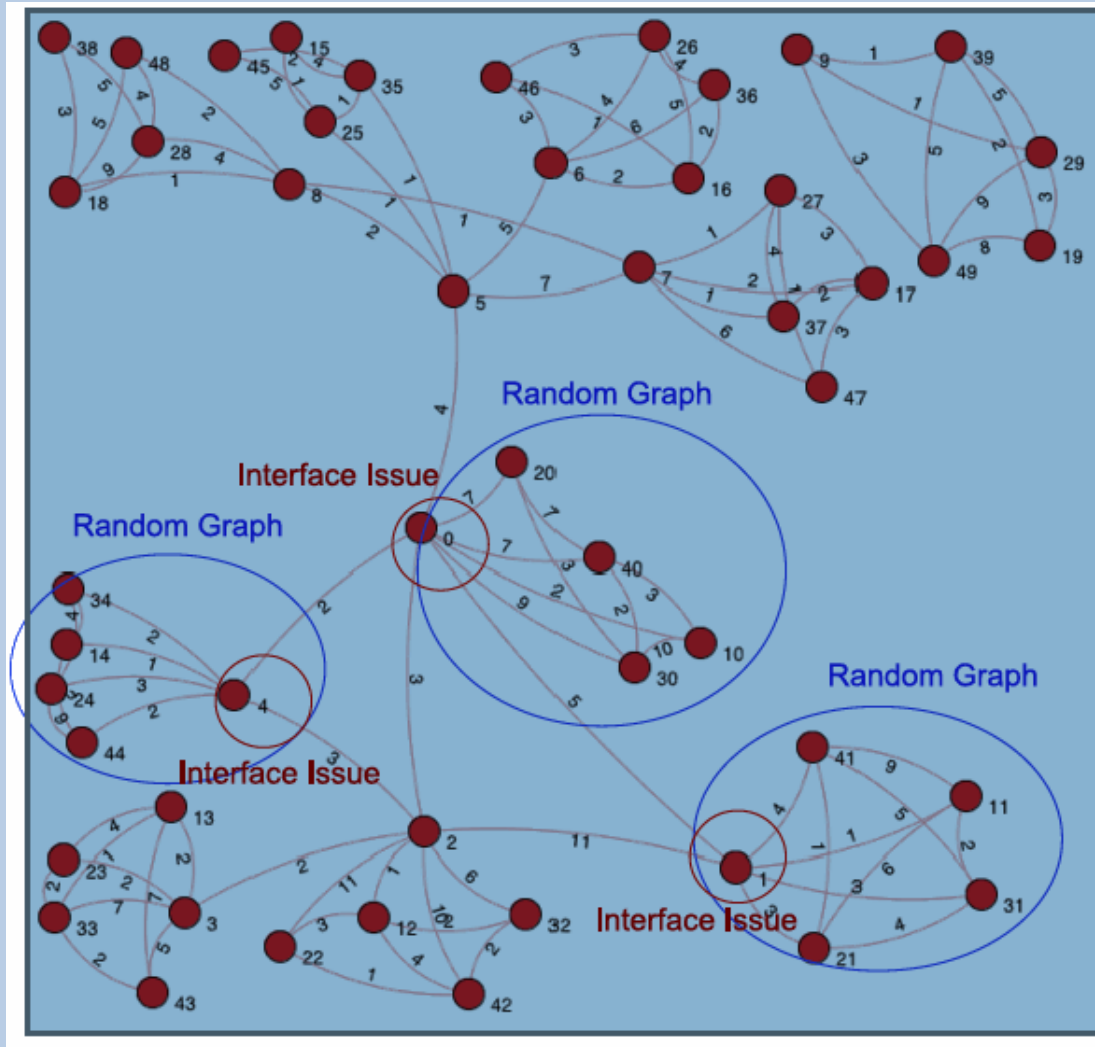
- 👍 Incentive compatible
- 👍 Good optimality
- 💬 **Lots** of ufun calculation and information revelation (2500 pair comparisons per **round**)

Constraint Auction Protocol



Minimizes info revelation & reduces computational complexity

Issue-Clumping



- 1) Negotiate within (almost) independent clusters
- 2) Combine sub-agreements
- 3) Negotiate to refine combined agreement

Hierarchical Agendas

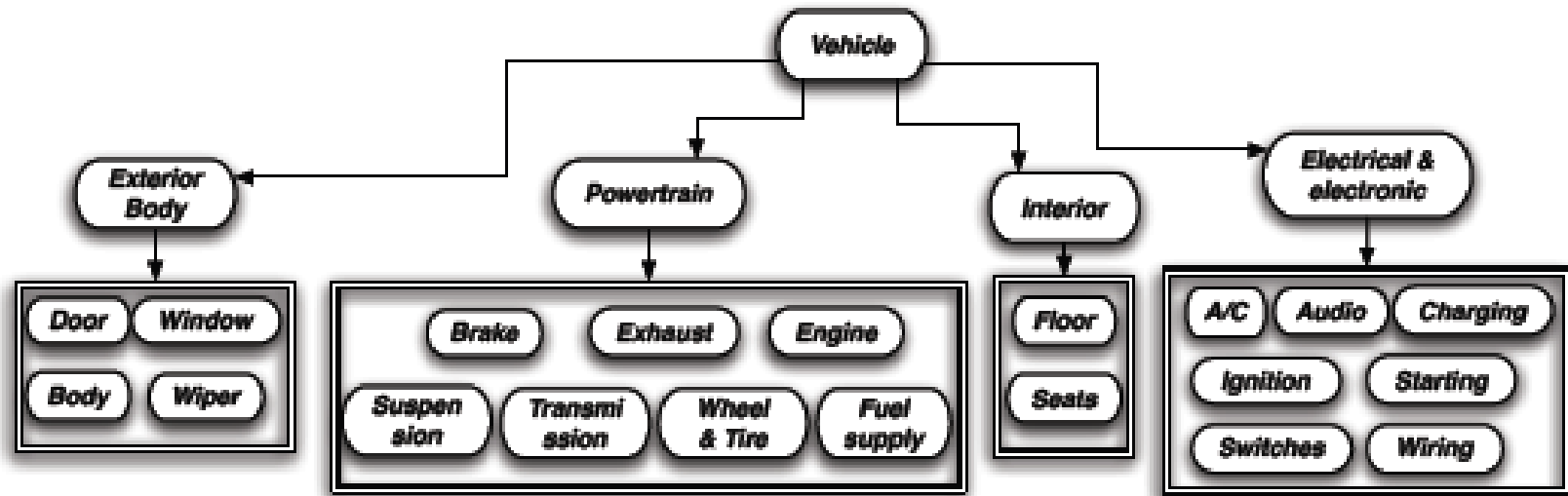


Figure 1: A hierarchical View of Car Design Process

Lessons Learned

- Complex decisions are everywhere
- Conventional (bargaining) negotiation techniques do poorly at this
- Solutions require innovative (and even counter-intuitive) techniques to achieve incentive compatibility and optimality at scale
- NegoWiki can help speed research in this area

Future Work

More scalable, people-friendly negotiation protocols

- Bag of lemons constraint auction
 - Better preference elicitation speed/accuracy
 - Fast convergence
- Truthful (limited) cooperative info exchange
 - Issue dependencies

Better incentive compatibility proofs

- e.g. based on bounded rationality

A Negotiation Handbook

NegoCANS

◆ Collaborators

- ◆ *Reyhan Aydogan (Delft University, Netherlands)*
- ◆ *Peyman Faratin (Robust Links, USA)*
- ◆ *Katsuhide Fujita (Tokyo University, Japan)*
- ◆ *Hiromitsu Hattori (Kyoto University, Japan)*
- ◆ *Takayuki Ito (Nagoya Institute of Technology, Japan)*
- ◆ *Catholijn Jonker (Delft University, Netherlands)*
- ◆ *Ivan Marsa-Maestre (University of Alcala, Spain)*
- ◆ *Rafik Hedfi (Nagoya Institute of Technology, Japan)*
- ◆ *Keisuke Hara (Nagoya Institute of Technology)*
- ◆ *Shantany Chakraborty (Nagoya Institute of Technology)*
- ◆ *Shelley Zhang (University of Massachusetts, USA)*

◆ Funding

- ◆ *National Science Foundation*

<http://cci.mit.edu/klein/>

Understanding the strengths and weaknesses of current social computing technology:

Emergence in Socio-
Technical Systems

Deliberation-Centric
Social Network Analysis

Developing new kinds of social computing that transcend the limitations of the old:

**Collective
Sensemaking**

*Understanding
complex problems*

Crowdmining

Deliberative Prediction
Markets

Learning by
Deliberation

**Collective
Innovation**

*Developing solutions
for complex problems*

The Deliberatorium

Artifact-Centric
Knowledge Sharing

Idea Filtering (Bag of
Lemons)

**Collective Decision
Making**

*Agreeing which
solutions to adopt*

Complex Negotiation

Conflict Management

**Collective
Action**

*Coordinating
solution enactment*

Robust Process Design

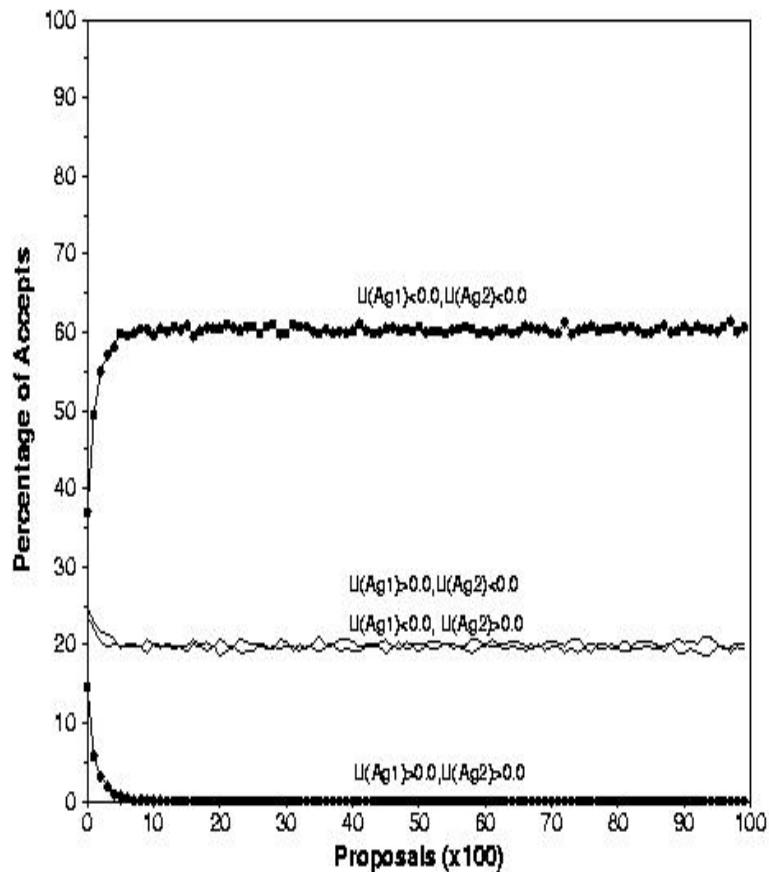
Exception Handling in
Distributed Systems

Pareto-Centric Decision-Making

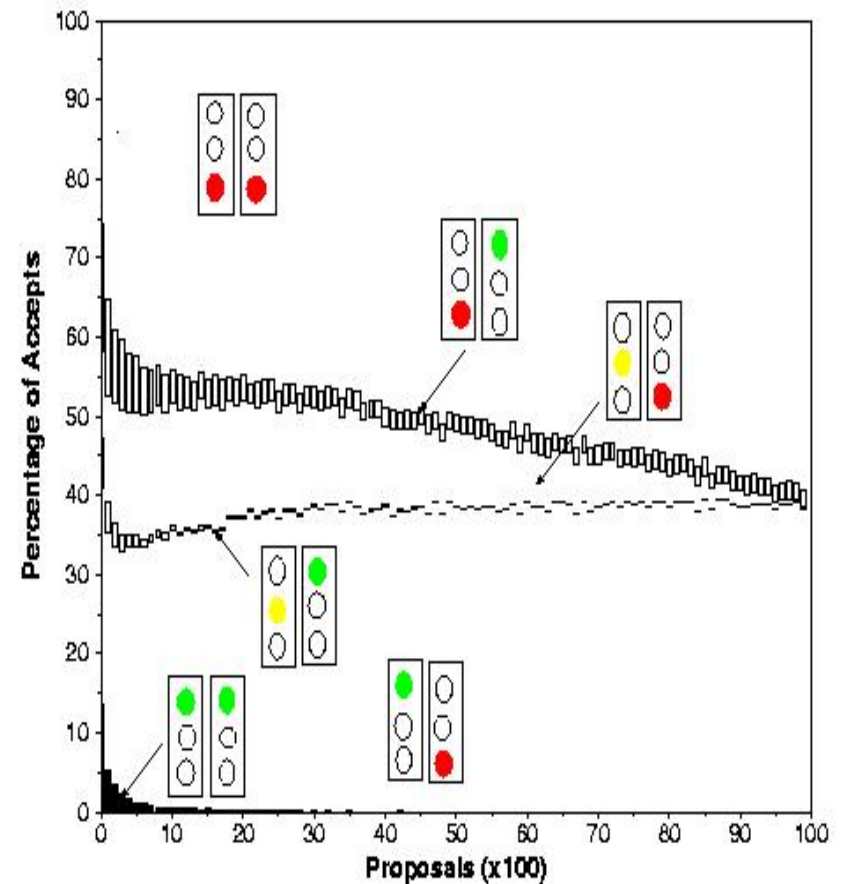
Mediated SA

Conflict Rates

2 Hillclimbers

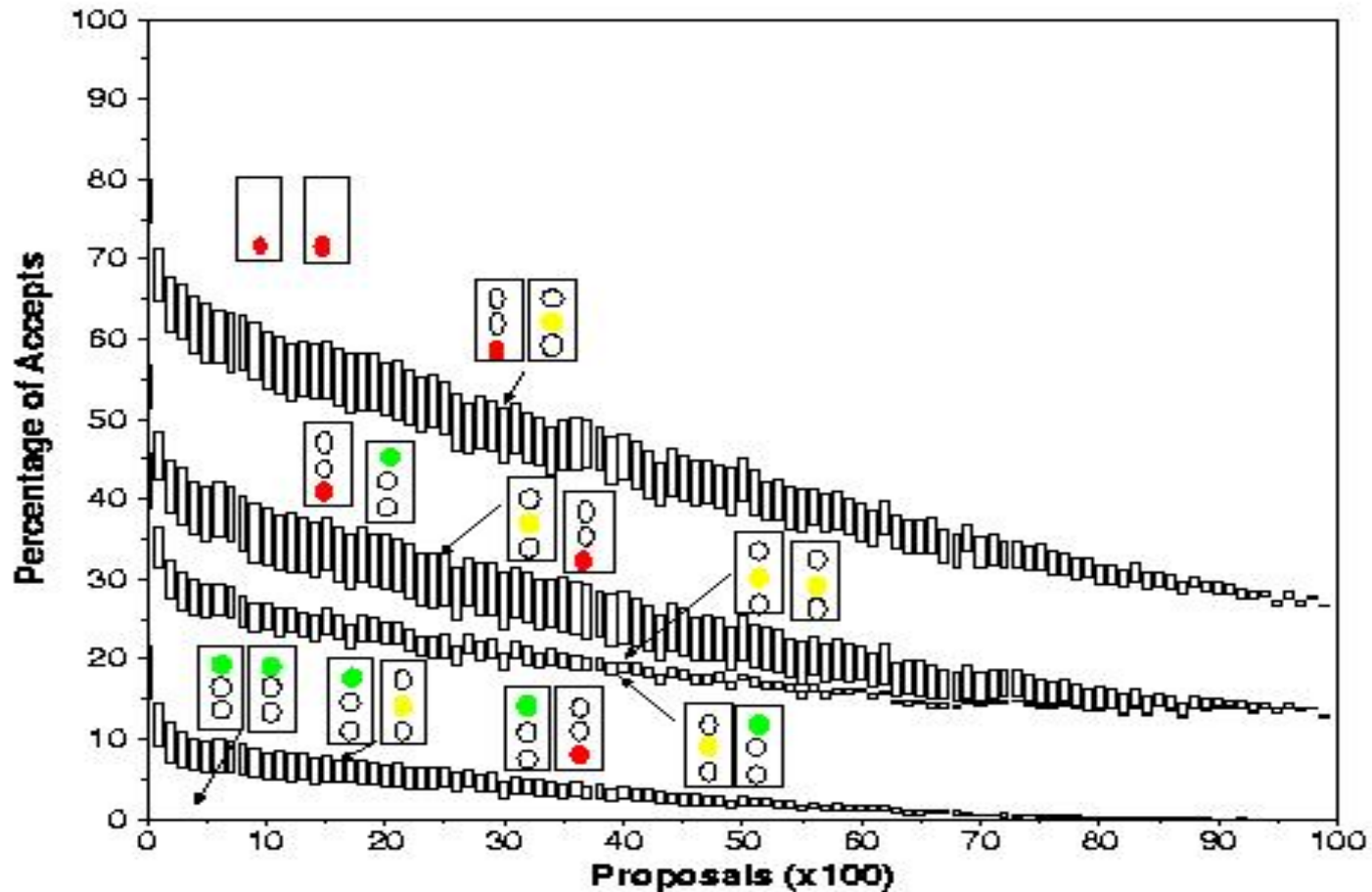


Assymmetric



Conflict Rate

2 Annealers



Intra-Negotiation Tit-for-Tat

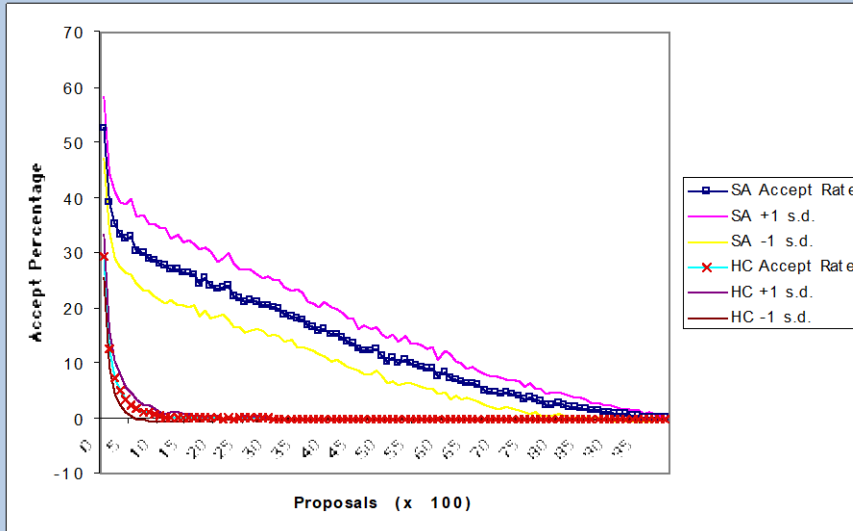
What if Agents start as cooperators

And switch to hill-climbers if the other hill-climbs
within a single multi-round negotiation

Could work if there are hundreds or thousands of rounds in a single negotiation

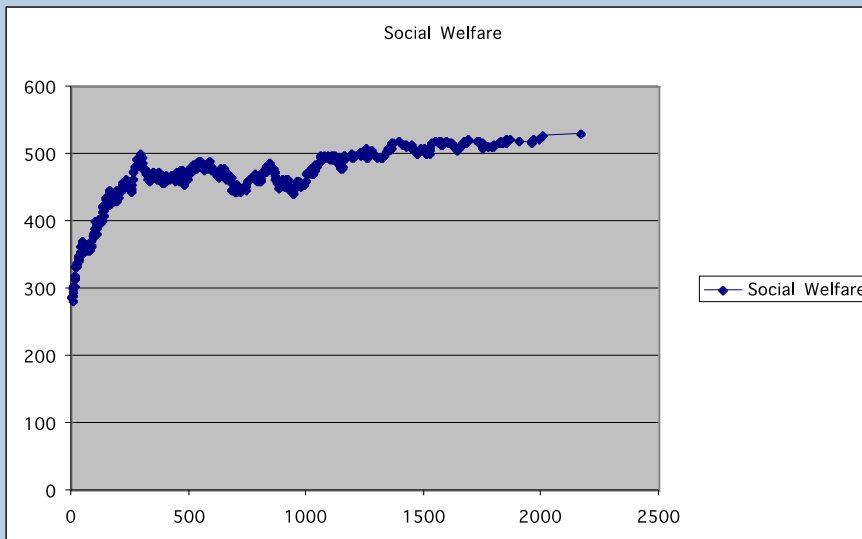
Detecting a Hill-Climber

Acceptance
Ratios



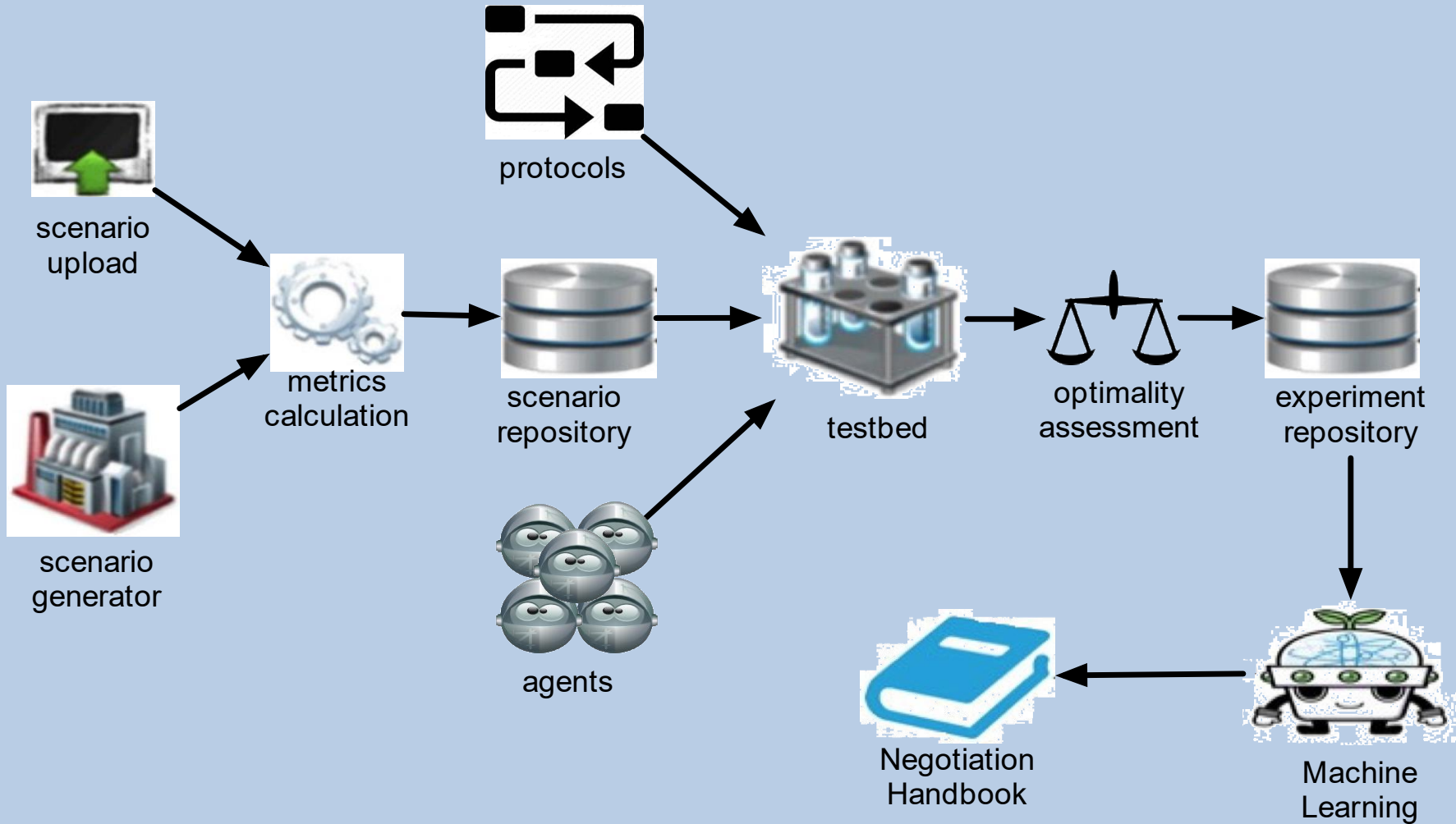
by the time hill-climbing can be detected, it's **too late** to avoid being a loser

Utility
Profile



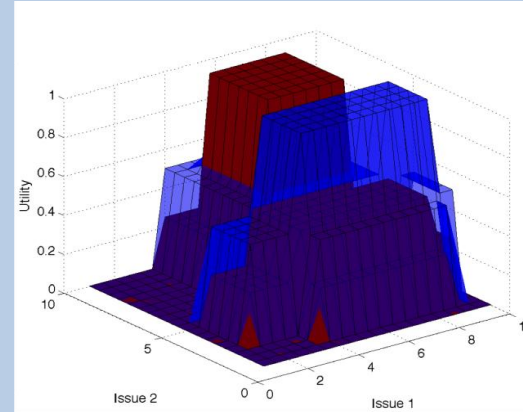
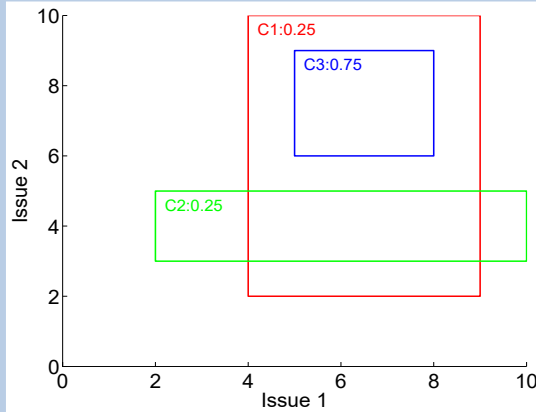
Evaluation Infrastructure

NegoWiki



Scenario Representation

Sum of *weighted hypervolumes*



Represented using **NegoXML**

```
<cube utility=10 >  
  <bound weight from 4 to 5>  
  <bound size from 3 to 10>  
</cube>
```

```
<plane slopes=(2.1 3.2) intercept=4>  
</plane>  
<bell height=12>  
</bell>
```

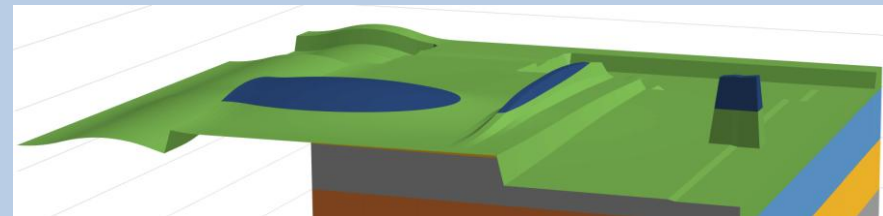
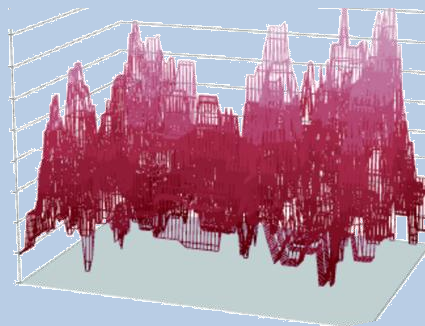
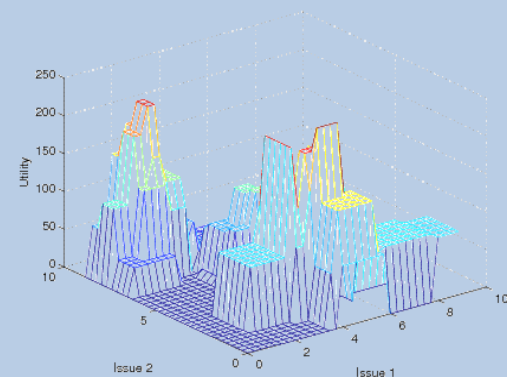
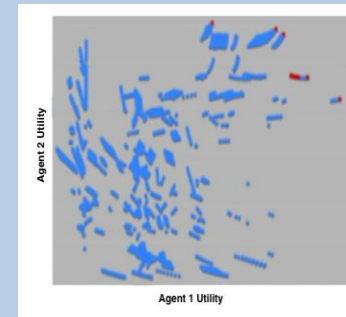
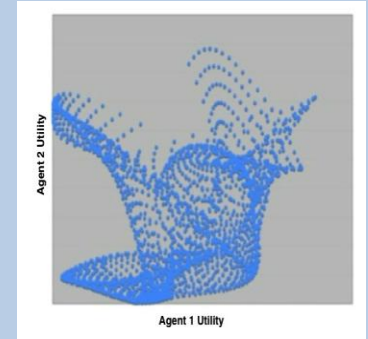
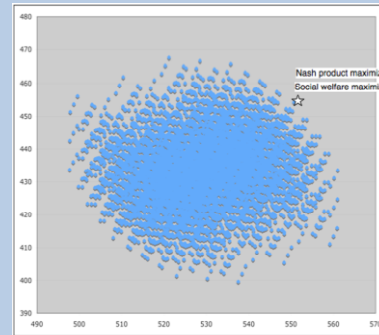
Subsumes all other nonlinear utility representations

Can **compress** utility functions => speed/size vs accuracy tradeoff



Scenario Generator

- Parameters
 - Domain (hierarchical)
 - # components
 - # issues
 - # agents
 - Constraints
 - # / type / size / dimensionality
 - # shared constraints across agents
 - fractal depth/breadth





Scenario Compiler

- Compiles hypervolume utility functions into decision trees that are *much* faster to run (e.g. 10 - 100x faster with 40 constraints)

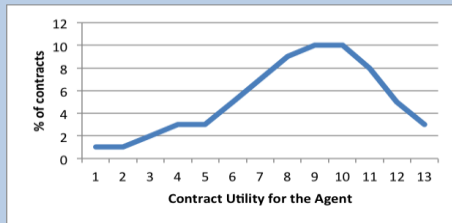
```
(SCENARIO :ID "S-1POQWMH-10"  
  :DOMAIN ((COMPONENT :NAME "c1"  
    :ISSUES ((ISSUE :INDEX 1 :TYPE INTEGER :NAME "c1-i2" :LEVEL 1 :MIN 0 :MAX 9)  
      (ISSUE :INDEX 2 :TYPE INTEGER :NAME "c1-i1" :LEVEL 1 :MIN 0 :MAX 9))))  
  :AGENTS ((AGENT :ID "A1"  
    :UFUN ((CUBE ((1 INCLUDES 2 9) (2 INCLUDES 0 1)) 52.0)  
      (CUBE ((1 INCLUDES 3 6) (2 INCLUDES 6 9)) 100.0)  
      (CUBE ((1 INCLUDES 0 9) (2 INCLUDES 1 2)) 70.0)  
      (CUBE ((1 INCLUDES 0 4) (2 INCLUDES 4 4)) 74.0))))))
```

```
(NODE :EDGE (1 9)  
  :LEFT (NODE :EDGE (1 6)  
    :LEFT (NODE :CONSTRAINTS ((CUBE ((1 INCLUDES 0 4) (2 INCLUDES 4 5)) 74.0)))  
    :RIGHT (NODE :CONSTRAINTS ((CUBE ((1 INCLUDES 3 6) (2 INCLUDES 6 9)) 100.0)))  
  :RIGHT (NODE :EDGE (2 2)  
    :LEFT (NODE :CONSTRAINTS ((CUBE ((1 INCLUDES 2 9) (2 INCLUDES 0 1)) 52.0)))  
    :RIGHT (NODE :CONSTRAINTS ((CUBE ((1 INCLUDES 0 9) (2 INCLUDES 1 2)) 70.0))))))
```

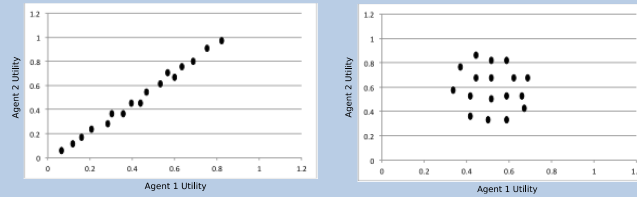


Scenario Metrics

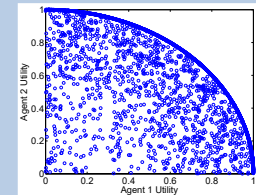
Utility Histogram



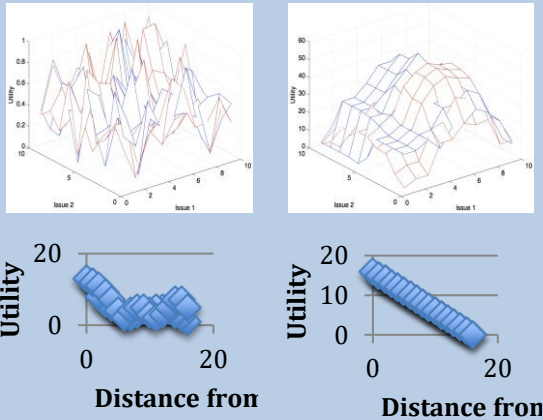
Cross-agent Correlation



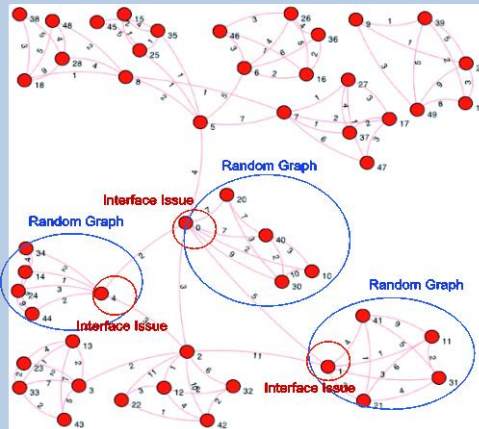
Pareto Distribution



Fitness Distance Correlation



Issue Dependency Graph





Scenario Archive

Source	# Agents	Domain
☒ Ito et al.,2007	☐ 2 ☐ 3 ☐ 4 ☐ 5	☐ 2x[0..9] ☐ 3x[0..9] ☐ 4x[0..9] ☐ 5x[0..9] ☐ 6x[0..9] ☐ 7x[0..9]

These constraints are satisfied by 116 test scenario(s)

[Download Scenarios](#)

Scenario: Ito et al 2007
[EDIT](#) [DOWNLOAD](#)

ID E-1M5UXGG-27
Agents 10
Issues 10
Domain 10
Description

Category: Ito et al.,2007

This scenario involves the kind of utility functions used in the paper by Ito et al *Multi-issue Negotiation Protocol for Agents: Exploring Nonlinear Utility Spaces*, published in *IJCAI 2007*^[1]. The paper used a *weighted_sum* of *plain utility functions* called constraints. The utility functions were generated as follows:

- Number of issues: 2 to 10.
- Domain for issue values: integers from 0 to 9.
- Constraint distribution: 5 unary constraints, 5 binary constraints, 5 trinary constraints, etc. (a unary constraint relates to one issue, a binary constraint relates to two issues, and so on).
- Maximum utility value for a constraint : $100 \times (\text{Number of Issues})$. Constraints that satisfy many issues thus have, on average, larger weights. This seems reasonable for many domains. In meeting scheduling, for example, higher order

Ito et al.,2007

Weighted sum of random plain utility functions

Summary

Weighted sum of random plain utility functions drawn from a uniform distribution

Key features

Number of issues	2,4,6,8,10
Domain of issues	$(x_1, x_2) x_i \in [0, 9] \subset \mathbb{R}$

Can be real problems, or from scenario generator (random w/parametrization)



Protocol Testbed

- Bilateral
 - Iterative concession
 - Iterative concession w/SA dice
- Mediated Multilateral
 - Simulated annealing (1 & 2 bit voting) w/tokens
 - *Propose-SA*
 - Genetic algorithms
 - Constraint auction
 - Pareto-concede
 - Tabu search
 - Veto
 - Single systematic
- Agenda management
 - Issue clustering (Fujita)
 - Hierarchical
 - *Agent clustering*



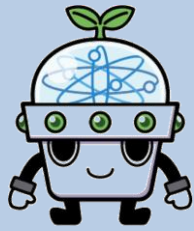
Experiment Repository

<u>Mechanism</u>	<u>Scenario</u>	<u>Contract</u>
SA v1	E-1M5UXGG-27	(31 16 4 40 1 12 45 35 13 10)
GA v2.1	E-1M5UXGG-27	(35 27 1 2 44 39 36 0 8 0)
CA v0.3	E-1M5UXGG-27	(25 40 17 29 43 40 49 11 48 1)
CA v0.3	E-1M5UXGG-29	(11 43 46 41 23 25 19 9 2 24)
HCA V4.1	E-1M5UXGG-29	(23 34 37 15 8 11 17 17 1 7)



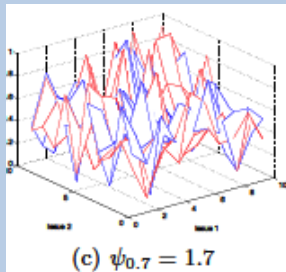
Optimality Assessment

- Social welfare & pareto optimality
 - Pareto front found using pareto-GA algorithm
- Fairness
- Negotiation time
- Information revelation
- Utility function calculation cost

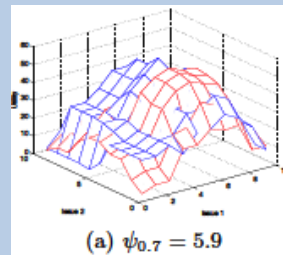


Mechanism Selection

Apply machine learning to correlations between scenario metrics and outcome optimality to find the *best* negotiation mechanisms for each negotiation type



Rugged, low
peak variance
=> Mediated
SA



Smooth, low agent
correlation => GA

Aydogan, R., Marsa-Maestre, I., Klein, M., & Jonker, C. M. (2015). *A Machine Learning Approach for Mechanism Selection in Complex Negotiations*. In Proceedings of The Eighth International Workshop on Agent-based Complex Automated Negotiations (ACAN 2015). Springer.

NegoCANS

Complex Adaptive Network Systems

Definition

- *Networks*: interlinked nodes for moving resources
- *Agents*: users adaptively optimizing their resource access
- *Goal*: maximize (weighted) social welfare of agents

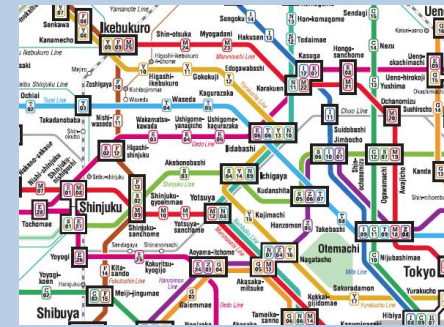
A ubiquitous real-world challenge

Why negotiate?

- *Load balancing*
- *Coordinated emergency response*

A **hugely** challenging computational problem

- *Scale*: 10^3 network nodes, 10^5 agents
- *Dynamics*: demands and availability change by second
- *Complexity*: high interdependency among CANS components
- *Self-interest*: agents demand privacy, prioritize own interests



Possible Solutions

Centralized optimization ✖

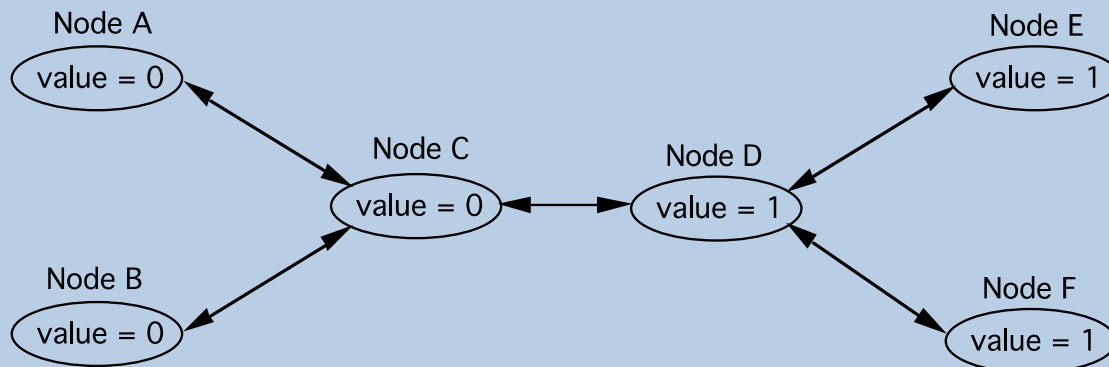
- Requires full preference revelation – privacy issue
- Unrealistic network/computation requirements; bottleneck

Combinatoric auctions ✖

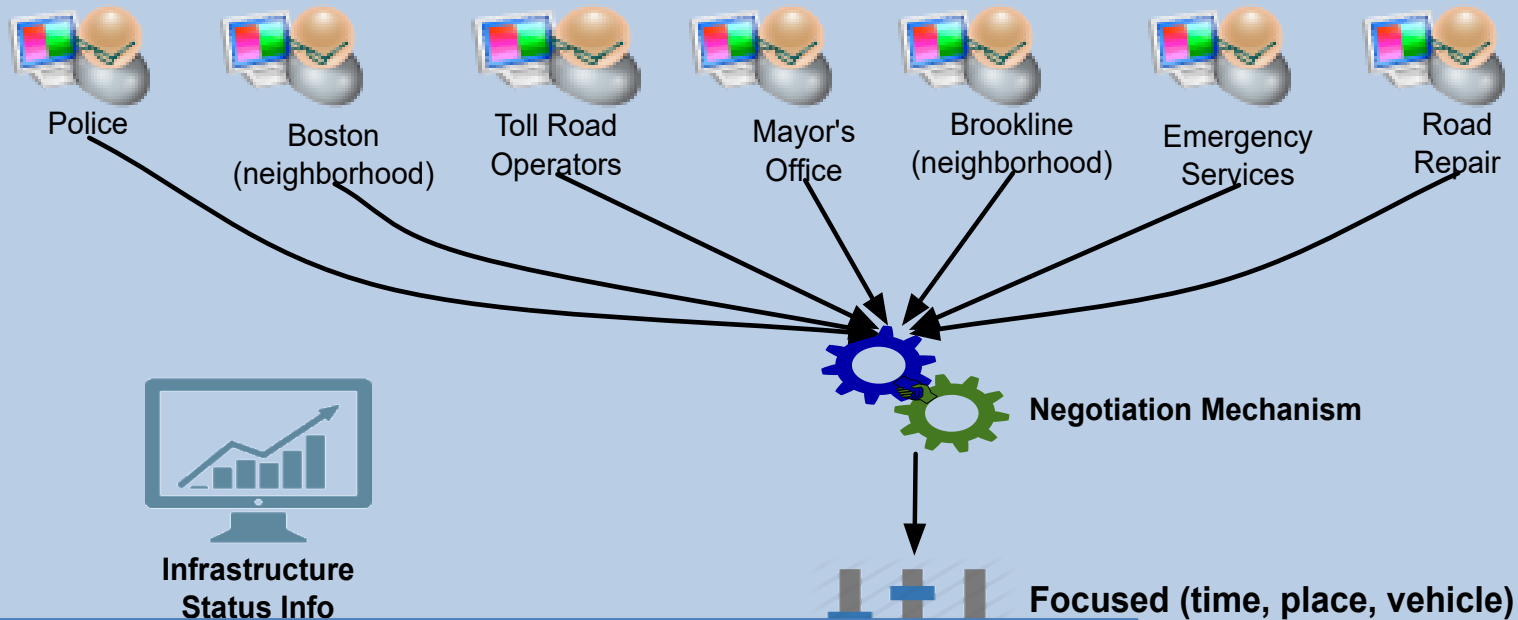
- Unrealistic bidding burden for agents
- Unrealistic network/computation requirements; bottleneck

P2P negotiation ✖

- Complexity for agents: negotiate with everyone?
- Emergence: local optimization $\neq$ global optima



NegoCANS ✓



Status changes, agent requests trigger re-negotiation

- e.g. I'll pay 50% more to save 30 minutes from X to Y
- e.g. traffic accident
- e.g. baseball game is ending
- e.g. emergency vehicles need expedited route

ves/constraints

s (incentives

and 6 am

between 6 and 8 am

ake resource utilization

decisions in this policy context
e.g. using "smart GPS"

Benefits

Scalability

- At each level, small # of negotiators and issues (10, not 100K)

Autonomy

- Resource users make their own decisions

Optimality

- Every agent contributes its intelligence **AND**
- Global context incorporated in all decisions

Privacy

- Resource users provide only requests (or nothing)

Adaptivity

- Can re-negotiate constraints as conditions change

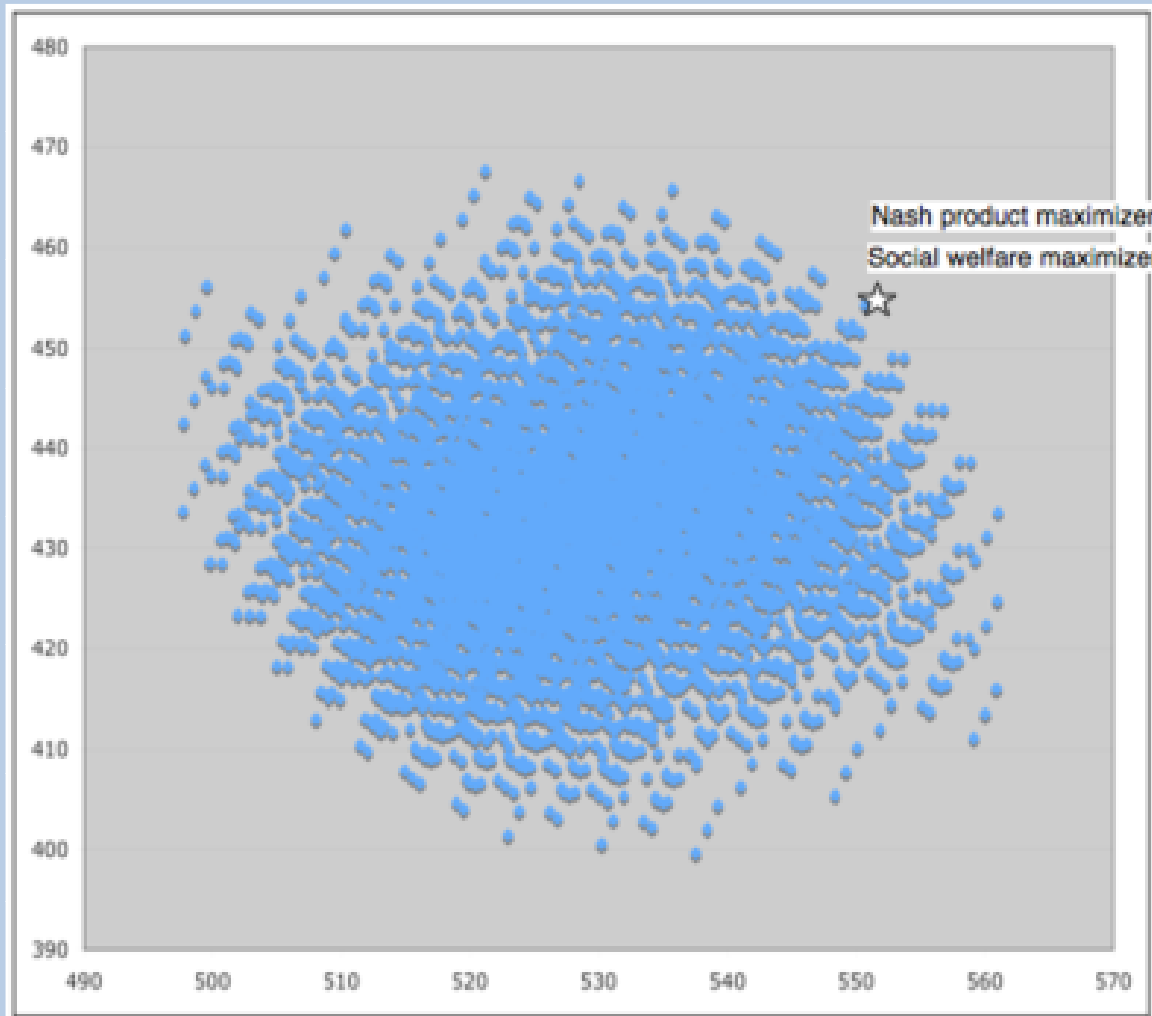
Practicality

- Doesn't require (but is improved by) giving agents "smart GPS"

Complexity even changes how we
assess agreements

Nash Product* with Simple Negotiations

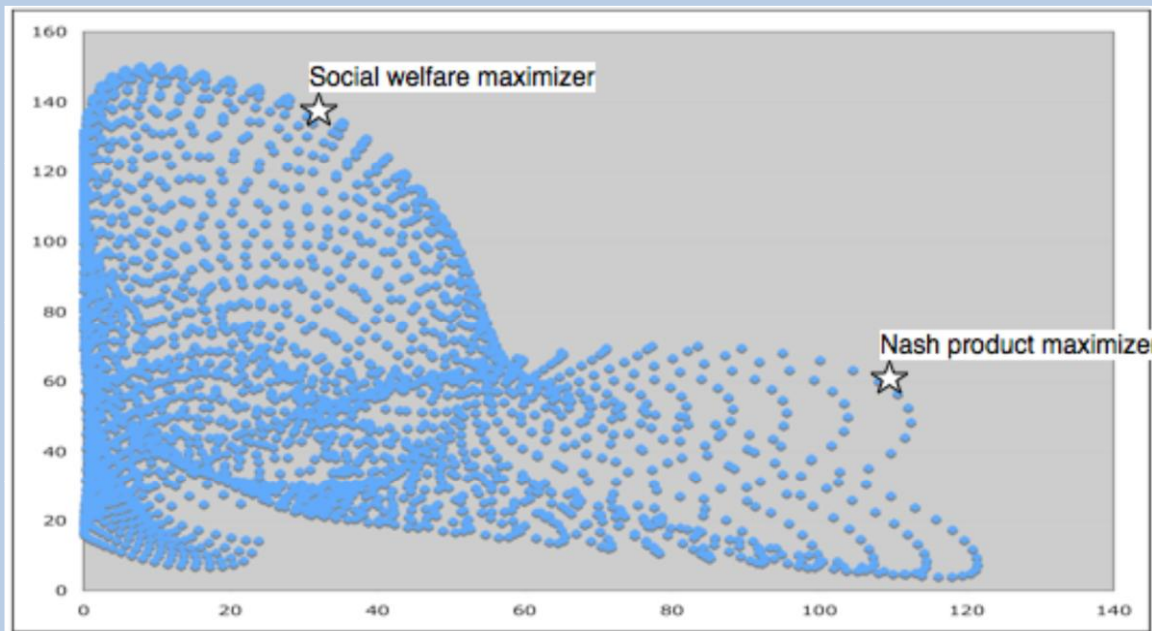
* Product of agent utilities



Simple contracts always produce convex pareto fronts

The Nash Product finds both *fair* & *efficient* contracts in this case

Nash Product with Complex Negotiations



Complex contracts can lead to complex *concave* pareto fronts

Fair & efficient can diverge from each other and even become ambiguous